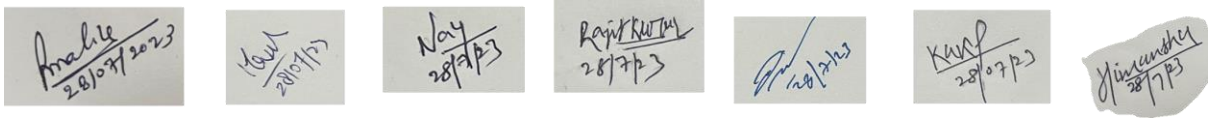
Syllabus

B.Sc. (Hons) Statistics & Data Science



1. Course Name	Descriptive Statistics	**L**	**T**		**P**	
2. Course Code	17200102	4	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (Use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This course prepares students to a rigorous study of descriptive statistics. Main topics are introduction to statistics and examples such as Measures of Central Tendency, Measures of dispersion, and correlation coefficient etc.

8. Course Objectives:

1. To develop the student's ability to deal with numerical and quantitative issues in business.
2. To enable the use of statistical, graphical and algebraic techniques wherever relevant.
3. To have a proper understanding of Statistical applications in Economics and Management.

9. Course Outcomes (COs):

After completing the course, students are expected to be able to:

1. Describe and discuss the key terminology, concepts tools and techniques used in statistical analysis
2. Critically evaluate the underlying assumptions of analysis tools
3. Discuss critically the uses and limitations of statistical analysis
4. Conduct basic statistical analysis of data.
5. Discuss the concept of Regression and Correlation.

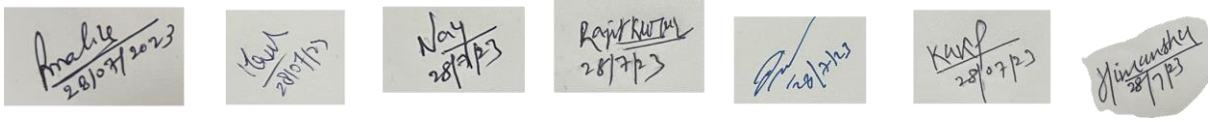
10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Introduction of data with graph**

Variable and data, types of variables, Types of Variables, Graphs for Categorical Data, Graphs for Quantitative Data, Pie Charts and Bar Charts, Line Charts, Dotplots, Stem and Leaf Plots, Relative Frequency Histograms.

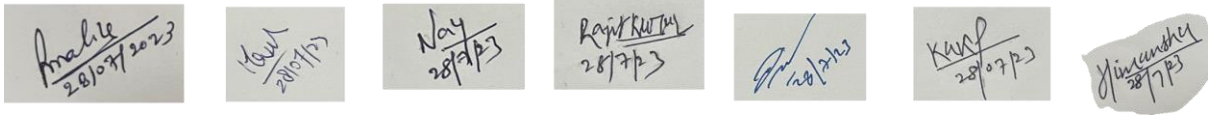
Unit – 2	**Number of lectures = 15**	**Title of the unit: Introduction to Statistics**

Concepts of Primary and Secondary data. Methods of collection and editing of primary data, Secondary data, Designing a questionnaire and a schedule, Measures of Central Tendency - Mean, Median, Mode, Geometric Mean and Harmonic Mean.



Unit – 3	Number of lectures = 10	Title of the unit: Measures of dispersion
Range, Quartile Deviation, Mean Deviation and Standard Deviation, Central and Non-Central moments and their interrelationship, Sheppard's correction for moments, Skewness and kurtosis.		
Unit – 4	**Number of lectures = 20**	**Title of the unit: Regression and Correlation**
Scatter Diagram, Product moment correlation coefficient and its properties, Spearman's Rank correlation, Concept of linear regression. Principle of least squares. Fitting a straight line by method of least squares, Relation between regression coefficients and correlation coefficient, Concept, and use of coefficient of determination.		
11. Brief Description of self-learning / E-learning component		
https://www.youtube.com/watch?v=BceFKnWh68Y https://onlinecourses.nptel.ac.in/noc22_mg87/preview https://archive.nptel.ac.in/courses/111/104/111104120/		
12. Books Recommended		

1. Mendenhall, William, Robert J. Beaver, Barbara M. Beaver, Introduction to Probability and Statistics, Cenage learning, 2019.
2. V.K.Kapoor and S. C.Gupta, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, New Delhi.
3. Goon AM, Gupta MK,Das Gupta B, Fundamentals of Statistics , Vol-I, the World Press Pvt.Ltd.
4. Sanjay Arora and BansiLal, New Mathematical Statistics: Satya Prakashan , New Delhi.
5. Hogg. Tanis. Rao. Probability and Statistical Inference. 7th edition. Pearson.



1. Name of the Department: Mathematics						
2. Course Name	Introduction to Data Science	**L**	**T**		**P**	
3. Course Code	17200103	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

8. Course Description:

This course covers some fundamental topics of data Science. In this course the students will be taught Databases for Data Science, Data Pre-processing and Feature Selection, and Data mining and Text Mining.

9. Course Objectives:

The objective of this course is to introduce some fundamental topics of data science like Data Munging, Feature Generation and Feature Selection, Feature Selection algorithms, Filters-Wrappers - Decision Trees -Random Forests which are directly relevant in some other papers of Data Science course.

10. Course Outcomes (COs):

By the end of this course, students should be able:

1. To demonstrate the knowledge databases for Data Science.
2. To enable for solving real-world problems in scientific domains using Feature Selection algorithms.
3. To apply concepts of Data analysis like Decision Trees, Random Forests etc.
4. To apply data pre-processing techniques.
5. To draw accurate and useful conclusions from a data analysis

11. Unit wise detailed content

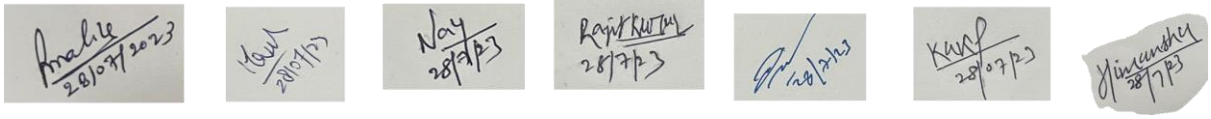
Unit-1	**Number of lectures = 15**	**Title of the unit: Introduction to data science**

What is Data Science? Need for data Science, data science process, Business Intelligence and Data Science, Prerequisites for a Data Scientist, Components of Data Science.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Databases for Data Science**

SQL – Tool for Data Science, Basic Statistics with SQL, Data Munging with SQL, MySQL, Filtering, Joins, and Aggregation, Window Functions and Ordered Data, Document Databases for Data Science, Graph Databases for Data Science.

Unit – 3	**Number of lectures = 15**	**Title of the unit: Data Pre-processing and Feature Selection**



Data cleaning, Data integration, Data Reduction, Data, Transformation and Data Discretization, Feature Generation and Feature Selection, Feature Selection algorithms, Filters, Wrappers, Decision Trees, Random Forests.

Unit – 4	**Number of lectures = 15**	**Title of the unit: Data Mining and Text Mining**

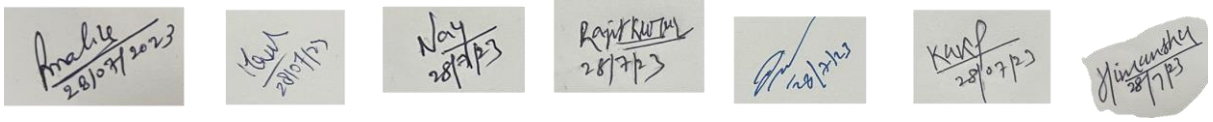
Data Mining Overview-Association Rule Mining-Text, Mining-Supervised and Unsupervised Learning, Supervised Learning via Support Vector Machines.

12. Brief Description of self-learning / E-learning component

https://nptel.ac.in/courses/106106179
https://www.digimat.in/nptel/courses/video/106107220/L01.html
https://www.youtube.com/watch?v=WEBUWYxaqLQ&list=PLD7HFcN7LXRcvobbHq_8zMyWq_tKwtebc

13. Books Recommended

1. Doing Data Science, Straight Talk from The Frontline, Cathy O'Neil and Rachel Schutt, O'Reilly (2014).
2. Data Mining: Concepts and Techniques", Third Edition, Jiawei Han, Micheline Kamber and Jian Pei, ISBN 0123814790, (2011).
3. Data mining methods,2nd edition, C. Rajan, Narosa (2016)
4. Jeffrey S. Saltz, Jeffre M. Stanton, "An Introduction to Data Science", Sage Publications (2017).
5. Nina Zumal and John Mount. Practical Data science in R, Managing Publication Company (2014).



1. Course Name	Calculus	L	T		P	
2. Course Code	17200105	2	0		0	
3. Type of Course (use tick mark)		Core (✓)	DSE ()	AEC ()	SEC ()	OE ()
4. Pre-requisite (if any)		5. Frequency (Use tick marks)	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	Tutorials = 0	Practical = 0

7. Course Description:

This course is designed to develop the topics of differential calculus. Emphasis is placed on limits, continuity, derivatives. Upon completion, students should be able to select and use appropriate models and techniques for finding solutions to derivative-related problems with and without technology, evaluation of areas, volumes of revolution, force, and work; fundamental formulas and various techniques of integration applied to both single variable and multi-variable functions.

8. Course Objectives:

Students that successfully complete this course will be able to:

1. Learn to find and use limits of functions,
2. To understand Rolle's Theorem, Mean Value theorems, series and Maclaurin's series.
3. Understand the meaning of differentiation and integration.
4. Apply the various methods of calculating derivatives of a function.
5. To discuss the application of integral in different areas.

9. Course Outcomes (COs):

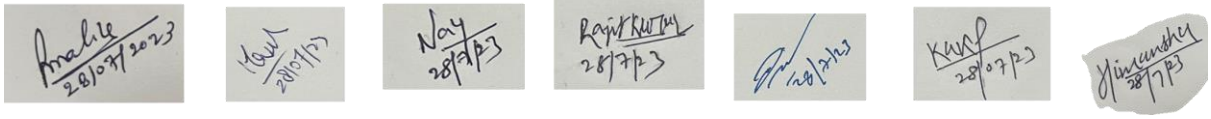
After completing the course, students are expected to be able:

1. To understand limits, continuity, and differentiability of functions in two variables and their properties.
2. To expand the two variables functions in series, form and discuss the maxima & minima using expansion of functions.
3. To understand the concept of asymptotes and curve tracing of the functions and their properties.
4. To evaluate the double & triple integral and apply the multiple integrals to find the area and volume.

10. Unit wise detailed content

Unit – 1	Number of lectures = 12	Title of the unit: Continuity and Differentiation

Limit and Continuity, Types of discontinuities, Differentiability of functions, Successive differentiation, n^{th} Differentiation of functions, Leibnitz's theorem, Limit, Continuity & Differentiability of functions in two variables, Partial differentiation, Euler's theorem on homogeneous functions and Jacobians.



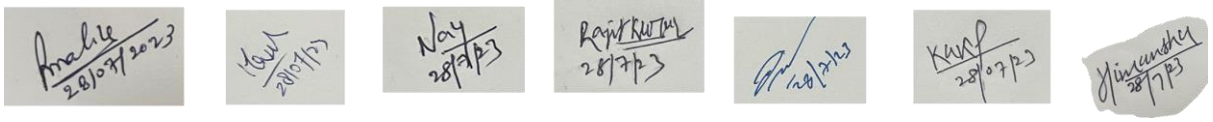
Unit – 2	Number of lectures = 05	Title of the unit: Mean Value Theorems and Series expansions
Rolle's theorem, Mean Value theorems, Taylor's theorem with Lagrange's and Cauchy's forms of remainder, Taylor's series, Maclaurin's series of $\sin x, \cos x, e^x, \log(1+x), (1+x)^m$, Taylor's theorem in two variables, Maxima and Minima of two variable functions,		
Unit – 3	**Number of lectures = 05**	**Title of the unit: Asymptote and Curve Tracing and Curvature**
Asymptotes in cartesian coordinates, Asymptotes in polar coordinates, Oblique asymptotes, Concavity, Convexity & Points of Inflexion, Tangents and normal curvature, Singular points, Tracing of curves in cartesian, Parametric and Polar coordinates.		
Unit – 4	**Number of lectures = 8**	**Title of the unit: Multiple Integrals and its Applications**
Double integral in cartesian and polar form, change of order of integration, Double integral by changing the variable, Area, Triple integral in cartesian and its application to find volume.		

11. Brief Description of self-learning / E-learning component

1. https://nptel.ac.in/courses/111/104/111104085/
2. https://nptel.ac.in/courses/111/107/111107108/

12. Books Recommended

1. Thomas and Finny, Calculus (9th ed.), Addison-Wesley, 1996.
2. Anton, Howard, Bivens, Irl, & Davis, Stephen, Calculus (10th ed.). John Wiley & Sons Singapore Pte. Ltd. Indian Reprint (2016).
3. D.V. Widder, Advanced Calculus (2nd ed.), Prentice Hall of India Pvt. Ltd, New Delhi, 2012.
4. Gorakh Prasad: Differential Calculus, Pothishasla Pvt. Ltd. Allahabad 2016.
5. Shanti Narayan And P. K. Mittal, Integral Calculus (35th ed.), S. Chand, New Delhi, 2005.



1. Name of the Department: Mathematics						
2. Course Name	Calculus Lab	**L**	**T**		**P**	
3. Course Code	17200105	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials = 0**		**Practical = 60**		

8. Course Description:

This course is designed to emphasize the knowledge of differential calculus using the visible picture of the function. Emphasis is placed on limits, continuity, derivatives, evaluation of areas, volumes of revolution, force, and work; fundamental formulas and various techniques of integration applied to both single variable and multi-variable functions.

9. Course Objectives:

Students that successfully complete this course will be able to:

1. To discuss the existence of limits and continuity of a functions with graph,
2. To discuss the existence of derivative of a function with graph,
3. Understand the meaning of differentiation and integration.
4. Apply the various methods of calculating derivative of a function.

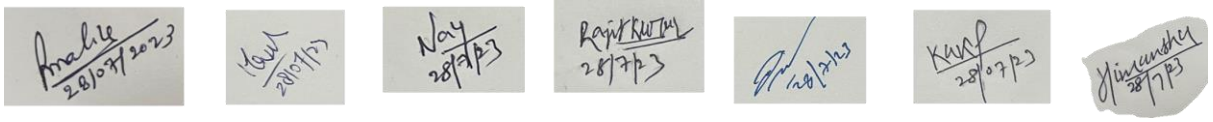
10. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To be able to evaluate various limit & continuity problem.
2. To use the concept of calculus an engineering and technology.
3. To carry out integration using the fundamental formulae and/or the various techniques of integration.
4. To demonstrate calculus models and their daily life application.
5. To Use the acquired skillsets in developing innovative projects for the benefit of society.

The list of practical's to perform in the computer lab.

1. To calculate limit, right hand limit and left-hand limit of functions and draw the graph of the function.
2. To calculate continuity of functions and draw the graph of the function.
3. To identify points of discontinuity with the help of graph
4. To calculate derivatives of a one and two variable functions with graph
5. To find Jacobians of the functions in two and three variables.
6. To discuss the applicability of Mean value theorems through graph
7. To calculate value of 'c' using Mean value theorems
8. To find Taylor's and Maclaurin's series
9. To find maximum and minimum of two variable functions.
10. Evaluate double and triple integrals.



11. Evaluate integrals using a change of variable.

12. Evaluate Area and volume

1. Gurpreet Singh Tuteja, “Practical Mathematics, International BOOK house Pvt Ltd.
2. https://www.mathworks.com/help/symbolic/solve-a-single-differential-equation.html
3. https://in.mathworks.com/help/symbolic/solve-a-system-of-differential-equations.html
4. https://www.mathworks.com/help/matlab/math/choose-an-ode-solver.html
5. http://www.math.tamu.edu/undergraduate/research/REU/comp/matode.pdf

11. Books Recommended

1. Shanti Narayan: Differential and Integral Calculus.
2. Murray R.Speigel: Theory and Problems of Advanced Calculus, Schaum’s Outline series, Schaum Publishing Co., New York.
3. N. Piskunov: Differential and Integral Calculus, Peace Publishers, Moscow.
4. Gorakh Prasad: Differential Calculus, Pothishasla Pvt. Ltd. Allahabad.
5. Gorakh Prasad: Integral Calculus, Pothishasla Pvt. Ltd. Allahabad.

1. Name of the Department: Mathematics						
2. Course Name	Analytics using Excel	**L**	**T**		**P**	
3. Course Code	17200101	0	0		4	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AEC ()**	**SEC (✓)**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials =0**	**Practical = 60**

8. Course Description:

This course provides all the tools necessary to create and use basic spreadsheets. Students will receive an overview of the data, logical functions, chart data techniques and learn the various methods for entering and editing data.

9. Course Objectives:

Learn the basics of creating spreadsheets in Excel. Enter data, formulas, format and print workbooks.

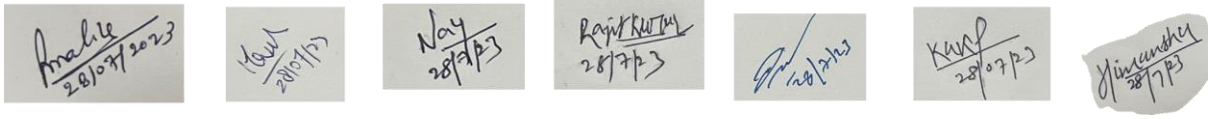
10. Course Outcomes (COs):

Upon completion of the course, students will be able to:

1. Use advanced functions and productivity tools to assist in developing worksheets
2. Manipulate data lists using Outline, Auto filter and PivotTables
3. Use Consolidation to summarize and report results from multiple worksheets
4. Use lookup function, Absolute Referencing Problems and Statistical functions.
5. Solve logical functions using statistical software.

11. Contents

1. Basic Functions: Sum / Average / Count / Max / Min/Basic Text / date / time / lookup
2. Data: Import from web/Import from text/Text to columns/Remove duplicates/Grouping and ungrouping
3. Absolute Referencing Problems with Absolute / Relative Cell Referencing/Creating Absolute
4. LOOKUP Functions
5. PivotTables: Creating, Formatting Simple PivotTables, Page Field in a PivotTable, formatting a PivotTable, Creating / Modifying a PivotChart.
6. Logical Functions: IFs and Nested IF Functions/Using AND / OR / NOT Functions
7. Statistical: Using the SUMIF / COUNTIF Functions/Using The AVERAGE / COUNT /



LARGER / SMALLER Functions.

8. Chart Data Techniques.
9. Formatting Worksheets: Formatting Numbers/Changing the Font Format/Aligning Cell Contents/Merging Cells/Adding Borders/Applying Cell Styles.

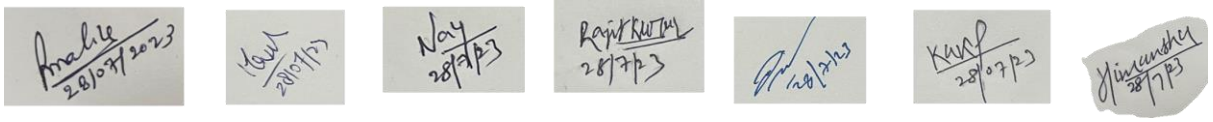
12. Web Links:

https://onlinecourses.nptel.ac.in/noc22_mg35/preview
https://www.youtube.com/watch?v=rduIHLghiLY

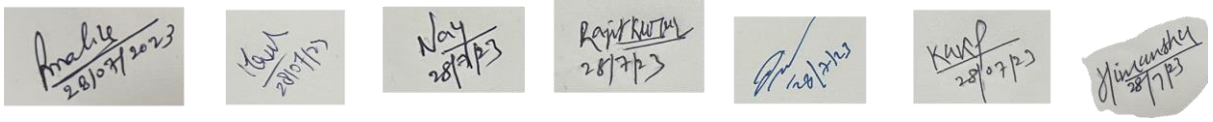
13. Books Recommended

1. Greg Harvey, Excel for Dummies, For Dummies, 2nd edition 1994.

2. John Walkenbach, Excel 2016, Wiley.



Semester-II



1. Course Name	Probability & Distribution Theory	L	T		P	
2. Course Code	17200202	4	0		0	
3. Type of Course (use tick mark)		Core (✓)	DSE ()	AEC ()	SEC ()	OE ()
4. Pre-requisite (if any)		5. Frequency (use tick marks)	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	Tutorials = 0	Practical = 0

7. Course Description:

This course provides a solid undergraduate foundation in both probability theory and mathematical probability. Topics include Basics in probability theory, random variables, expectation and variance, special probability distributions.

8. Course Objectives:

1. To develop the skills of the students in the area of Probability and Statistics
2. To expose the students to the basics of probability distributions and application of family of random variables in real life situations
3. To understand basic concepts in probability theory, learn commonly used probability distributions.

9. Course Outcomes (COs):

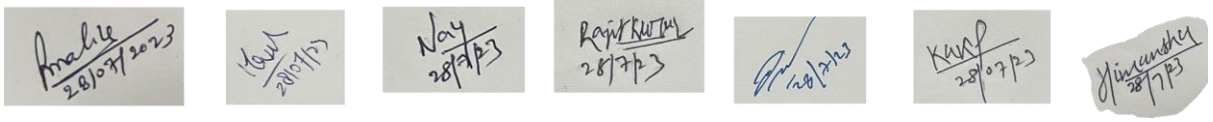
After completing the course, students are expected to be able:

1. To enable to the basic concept of theory of probability.
2. To acquire knowledge of central limit theorem and its application.
3. To know about Expectation and generating function.
4. To understand the concept of Binomial, Poisson and Normal distributions.
5. To know about continuous and discrete probability and distribution and their application.

10. Unit wise detailed content

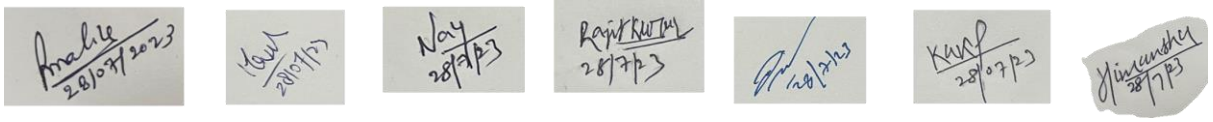
Unit – 1	Number of lectures = 15	Title of the unit: Introduction to Probability

Basic Concepts of Probability, random experiments, trial, outcome, sample space, event, mutually exclusive and exhaustive events, equally likely and favorable outcomes, Mathematical, Statistical, axiomatic definitions of probability. Conditional Probability and independence of events, Bayes' theorem.



Unit – 2	Number of lectures = 15	Title of the unit: Random variable
Definition of random variable, discrete and continuous random variables, functions of random variable. Probability mass function, probability density function, Distribution function and its properties, bivariate random variable - meaning, joint, marginal and conditional Distributions, independence of random variables.		
Unit – 3	**Number of lectures = 15**	**Title of the unit: Expectation and generating Function**
Mathematical expectation, Addition and Multiplication theorem of expectation, Moment generating functions, Expected value, Law of large numbers, The Variance and Standard Deviation, Some Theorems on Variance Standardized Random Variables, Moments Moment Generating Functions, Some Theorems on Moment Generating Functions.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Discrete and Continuous Probability Distributions**
Discrete and Continuous probability distributions: Binomial, Poisson and Normal distributions, Some Properties of the Normal Distribution, Relation Between Binomial and Normal Distributions, Some Properties of the Poisson Distribution, Relation Between the Binomial and Poisson Distributions, Relation Between the Poisson and Normal Distributions, The Central Limit Theorem.		
11. Brief Description of self-learning / E-learning component		
https://archive.nptel.ac.in/courses/111/102/111102111/ https://archive.nptel.ac.in/courses/111/104/111104079/ https://www.youtube.com/watch?v=O-YhNpy7z3I		
12. Books Recommended		

1. Robert V. Hogg, Joseph W. McKean and Allen T. Craig, *Introduction to Mathematical Statistics*, Pearson Education, Asia, 2007.
2. Myer, P.L. (1970): Introductory Probability and Statistical Applications, Oxford & IBH Publishing, New Delhi
3. Gupta, S.C. and V.K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, New Delhi, 2008.
4. Richard Arnold Johnson, Irwin Miller, John E. Freund, Miller and Freund's Probability and Statistics for Engineers, 8 illustrated, Prentice Hall, 2011
5. William Feller, An introduction to probability theory and its applications, Vol. I, 3rd edition, John Wiley & Sons.



1. Course Name	Linear Algebra	**L**	**T**		**P**	
2. Course Code	17200203	4	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

An introduction to linear algebra and how it can be used, including basic mathematical proofs. Topics include systems of equations, vectors, matrices, orthogonality, subspaces, and the eigen value problem

8. Course Objectives:

Problems in linear algebra arise in a wide variety of scientific and engineering applications including the design of structures, the analysis of electrical networks, and the modeling of chemical processes. This course will cover the analysis and implementation of algorithms used to solve linear algebra problems in practice

9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To demonstrate about basic knowledge of vector space
2. To apply linear transformations & their properties for solving mathematical and computational problems such as computer graphics.
3. To analyze mathematical problems using Jordan canonical form.
4. To discuss well-known research problems regarding Inner product and Diagonalization.
5. To know about continuous and discrete probability and distribution and their application.

10. Unit wise detailed content

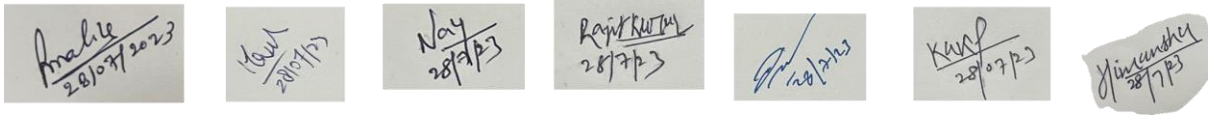
Unit – 1	**Number of lectures = 15**	**Title of the unit: Matrices and Determinants**

Matrices: Elementary transformations, Echelon and normal form, rank, consistency of system of linear equations, solution of homogeneous and non-homogeneous systems of linear equations, Gauss Elimination Method, Gauss-Jordan Method

Unit – 2	**Number of lectures = 15**	**Title of the unit: Vector spaces**

Vector spaces: Fields, Vector spaces over a field, subspaces, Linear independence and dependence, existence of basis, coordinates, dimension.

Unit – 3	**Number of lectures = 15**	**Title of the unit: Linear Transformation**



Linear Transformations: Rank Nullity Theorem, isomorphism, matrix representation of linear transformation, change of basis, similar matrices, linear functional and dual space.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Inner product spaces and Diagonalization**
Gram-Schmidt orthonormalization, orthonormal basis, orthogonal projection, projection theorem, Diagonalization: Eigenvalues and eigenvectors, diagonalizability, Invariant subspaces, adjoint of an operator, normal, unitary and self-adjoint operators.		
11. Brief Description of self-learning / E-learning component		
1. http://home.iitk.ac.in/~arlal/book/nptel/pdf/booklinear.html 2. http://www.maths.qmul.ac.uk/~pjc/notes/linalg.pdf 3. http://www.mathe2.uni-bayreuth.de/stoll/lecture-notes/LinearAlgebraI.pdf		
12. Books Recommended		
1. Kenneth Hoffman and Ray Kunze: Linear Algebra, PHI publication. 2. Gilbert Strang: Linear Algebra and Its Applications, 4th edition. 3. Sheldon Axler: Linear Algebra Done Right, UTM, Springer.		

1. Name of the Department: Mathematics						
2. Course Name	Differential Equations	**L**	**T**		**P**	
3. Course Code	17200204	2	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 30		**Tutorials = 0**		**Practical = 0**		

8. Course Description:

Differential equations and their solutions. Linear Differential equations. Homogeneous Differential Equations. Second order linear differential equations. Total differential equations.

9. Course Objectives:

To introduce the basic concept of Differential equations and their solutions. Strength of these concepts in engineering and real-world problems will be highlighted.

10. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To learn about the basic concept of Differential equations.
2. To understand orthogonal trajectory and Homogeneous ordinary differential equation.
3. To learn about the basic concept of second order Differential equations.
4. To solve the Differential equation by general method.
5. To solve the Method of variations of parameters.

11. Unit wise detailed content

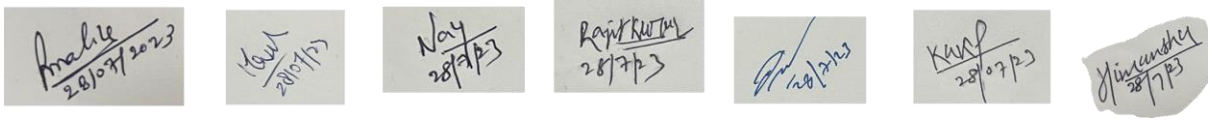
Unit-1	**Number of lectures = 8**	**Title of the Unit: Introduction to Differential Equations**

Order and degree of a differential equation, Linear & Non-Linear differential equations, Homogeneous differential equations, Geometrical meaning of differential equations (DE), Exact differential equations, Integrating factors, Clairaut's equations, Singular solutions of differential equations.

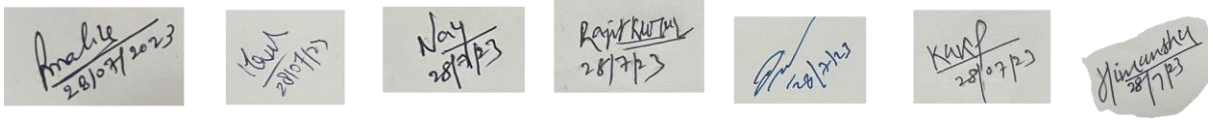
Unit - 2	**Number of lectures = 10**	**Title of the unit: Orthogonal Trajectories and Homogeneous DE**

Orthogonal trajectories in cartesian coordinates and polar coordinates, Self-orthogonal family of curves, Linear differential equations with constant coefficients, Solution of Homogeneous & Non-Homogeneous linear differential equations, Cauchy's differential equations.

Unit – 3	**Number of lectures = 8**	**Title of the unit: Linear DE and Non-homogeneous DE**



Linear DE of second order with variable coefficients: When a part of CF is known, Reduction to normal form, Transformation of the equation by changing the dependent variable/independent variable, Method of variations of parameters		
Unit - 4	**Number of lectures = 4**	**Title of the unit: Solutions of Simultaneous DE**
Ordinary simultaneous DE, Solution of simultaneous DE involving operators x (d/dx) or t (d/dt) etc, Simultaneous equation of the form dx/P=dy/Q=dz/R.		
12. E-learning component		
1. https://nptel.ac.in/courses/111/106/111106100/ 2. https://nptel.ac.in/courses/111/107/111107111/		
13. Books Recommended		
1. A.R. Forsyth, A Treatise on Differential Equations, Macmillan and Co. Ltd. London (1956). 2. E.A. Codington: An Introduction to Differential Equations, Dover (1961). 3. S.L. Ross: Differential Equations (3rd), John Wiley & Sons (2004). 4. B. Rai and D.P. Chaudhary: Ordinary Differential Equations: An Introduction; Narosa, Publishing House Pvt. Ltd, (2019).		



1. Name of the Department: Mathematics						
2. Course Name	Differential Equations Lab	**L**	**T**		**P**	
3. Course Code	17200205	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials = 0**	**Practical = 60**

8. Course Description:

This course is designed to emphasize the knowledge of differential equations. Emphasis is placed on different forms of linear and non-linear differential equations. Upon completion, students should be able to write the programs in MATLAB or other software's.

9. Course Objectives:

1. Give an account of basic concepts and definitions for differential equations.
2. Use methods for obtaining exact solutions of linear homogeneous and non-homogeneous differential equations.
3. Describe some simple numerical solution techniques and be familiar with mathematical software for differential equations.
4. Use elementary methods for linear systems of differential equations.

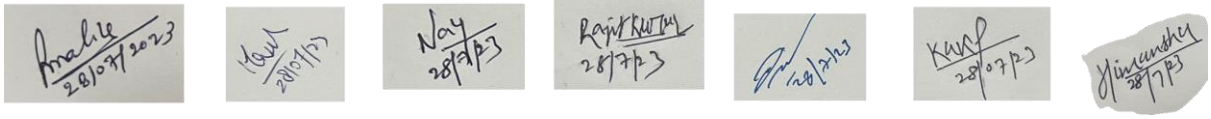
10. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To introduce the students to the technique of solving various problems of engineering and science
2. To solve first-order ordinary differential equations using MATLAB.
3. To learn about the explicit solution of first-order differential equations
4. To Solve basic application problems described by second-order linear differential equations with constant coefficients.
5. To solve differential equations analytically

11. The list of practical's to perform in the computer lab.

1. Solve a differential equation analytically by using the dsolve function, with or without initial conditions.
2. To solve a system of differential equations,
3. To solve first order Linear ODE
4. To solve First order linear differential equations
5. To solve first order Bernoulli equations
6. To solve ODE with initial conditions
7. To solve Nonlinear differential equations with initial conditions



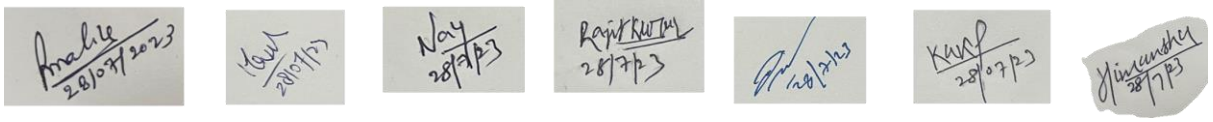
8. To solve second order ODE with initial conditions
9. To solve a system of Ordinary differential equations
10. To find the explicit solution of first order differential equations
11. To find the explicit solution of second order differential equations

12. E-learning component

1. https://www.mathworks.com/help/symbolic/solve-a-single-differential-equation.html
2. https://in.mathworks.com/help/symbolic/solve-a-system-of-differential-equations.html
3. https://reference.wolfram.com/language/tutorial/DSolveIntroduction.html

13. Books Recommended

1. Barnes, Belinda &Fulford, Glenn R.. Mathematical Modelling with Case Studies, Using Maple and MATLAB (3rd ed.). CRC Press, Taylor & Francis Group, (2015).
2. C. P.López, MATLAB Differential Equations, Apress, (2014).
3. Dingyü Xue, Differential Equation Solutions with MATLAB, De Gruyter, (2020).
4. Brian R. Hunt, Ronald L. Lipsman, John E. Osborn, Donald A. Outing, Jonathan Rosenberg, Differential Equations with Mathematica (3rd), John Wiley & Sons, (2009).



1. Name of the Department: Mathematics						
2. Course Name	Introduction to R Programming	**L**	**T**		**P**	
3. Course Code	17200201	0	0		4	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AEC ()**	**SEC (✓)**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials = 0**		**Practical = 60**		

8. Course Description:

This course will help students to gain a basic understanding of R programming. This course will cover basic concepts and techniques in R programming such as data in a file, Generation data, manipulating objects, Calculations with R Software, Arithmetic and Simple function, iterations, manipulating data and creating plots. You will learn base of R Programming and a few selected packages.

9. Course Objectives:

To introduce the basic concept of R programming. Strength of these concepts in engineering, data Science, Machine learning and real-world problems will be highlighted.

10. Course Outcomes (COs):

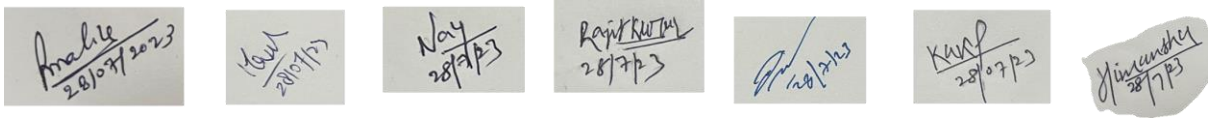
After completing the course, students are expected to be able:

1. To explain critical R programming concepts
2. To learn about the basic programming concepts such as data types, data structures and indexing, and use them in your work.
3. To analyze data and generate reports based on the data.
4. To solve descriptive Statistics using R.
5. To learn about the manipulation of data.

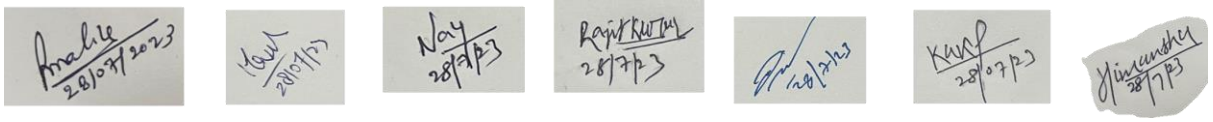
11. Unit wise detailed content

Unit-1	**Number of lectures = 30**	**Title of the Unit: Introduction to R Programming**

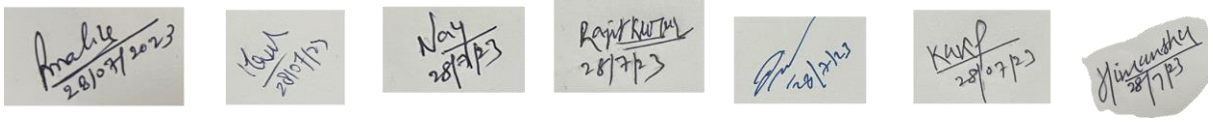
Introduction of R, How R works, Creating, listing and deleting the objects in memory, Objects, Reading and Writing data in a file, Generation data, regular sequences, Random Sequence, Manipulating objects, Calculations with R Software, Arithmetic and Simple function, Matrix computation.



Unit - 2	Number of lectures = 30	Title of the unit: Statistical Analysis with R
Graphics and Plots, Introduction to descriptive Statistics, central tendency of data, frequency distribution, generic function, discrete probability distribution, continuous probability distribution.		
12. E-learning component		
1. https://archive.nptel.ac.in/courses/111/104/111104146/ 2. https://search.r-project.org/CRAN/refmans/insight/html/find_statistic.html		
13. Books Recommended		
1. Wickham, H. and Grolemund G. (2017). R for Data Science, O′Reilly, 1st edition. 2. Akritas, M. (2016). Probability & Statistics with R for Engineers and Scientists, Pearson Education, 1st edition. 3. Grolemund, G. (2014). Hands-on programming with R: Write your own functions and simulations. Sebastopol, CA: O'Reilly Media.		



Semester-III



1. Course Name	Time Series Analysis	**L**	**T**		**P**	
2. Course Code	17200302	4	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical		
Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

The course will provide a basic introduction to time series analysis. We will cover time series components and exploratory data analysis, AR/MA models, model identification/estimation/linear operators, linear function, Exponential smoothing methods etc.

8. Course Objectives:

1. Learn basic analysis of time series data
2. learn basic concepts of index numbers
3. learn auto regressive and model averaging models
4. learn basic concepts of Exponential smoothing methods

9. Course Outcomes (COs):

After completing the course, students are expected to be able to:

1. Explain and interpret AR/MA models.
2. Explain and interpret index numbers.
3. Define different time series components.
4. Explain and interpret Box-Jenkins models.
5. Run and interpret time series models and Exponential smoothing models for time series.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Introduction to Time Series**

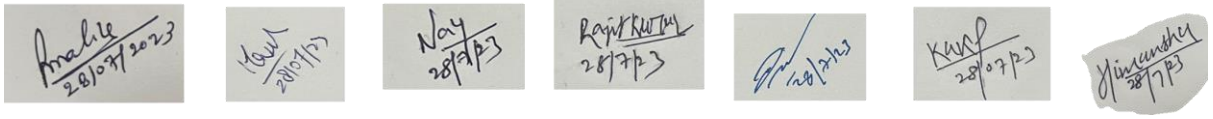
Time series definition, components of time series, Additive and multiplicative models, freehand curve, moving average, least square method.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Introduction to Trend**

Application of time series from various fields, Decomposition of time series, Measurement of trend, method of semi averages, fitting various mathematical curves, and growth curves.

Unit – 3	**Number of lectures = 15**	**Title of the unit: Box-Jenkins modelling**

Moving-average (MA) process and Autoregressive (AR) process of orders one and two. ACF and its graphical use in guessing the order of MA processes. Estimation of the parameters of AR (1) and AR (2) using least square and Yule-Walker equations, Exponential smoothing forecasting methods.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Index numbers**
Meaning and use of index numbers, Construction of index numbers, simple and weighted Index numbers, index numbers, Laspeyer's, Paasche's Marshall, Edgeworth and Fisher's index number, chain base and fixed base index number, construction of cost-of-living index number.		
11. Brief Description of self-learning / E-learning component		
https://archive.nptel.ac.in/noc/courses/noc22/SEM1/noc22-ch19/ https://nptel.ac.in/courses/109104182 https://nptel.ac.in/courses/109104182		
12. Books Recommended		
1. Chatfield C. (1980): The Analysis of Time Series –An Introduction, Chapman & Hall. 2. Kendall M.G. (1976): Time Series, Charles Griffin. 3. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2002): Fundamentals of Statistics, Vol. II, 8th Edn. The World Press, Kolkata 4. Allen R.G.D. (1975): Index Numbers in Theory and Practice, Macmillan 5. Shumway & Stoffer (2011) Time Series Analysis and its applications, with examples in R, 3rd edition, Springer.		



1. Name of the Department: Mathematics						
2. Course Name	Time Series Analysis Lab	**L**	**T**		**P**	
3. Course Code	17200303	2	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials =0**	**Practical = 60**

8. Course Description:

The aim of this course is to provide students with the essential expertise to handle time series techniques. We will cover time series components and exploratory data analysis, AR/MA models, model identification/estimation/linear operators, linear function, Exponential smoothing methods etc.

9. Course Objectives:

Make the students understand basics of Data Visualization, implement index numbers and write programs on visualization of time series, proportions.

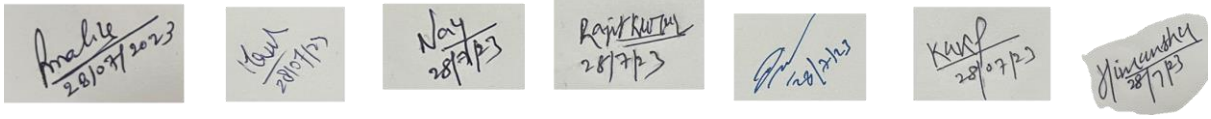
10. Course Outcomes (COs):

After completion of this course, the students will know about:

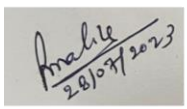
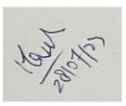
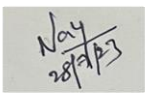
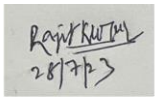
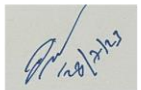
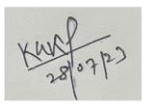
1. time series data, its applications to various fields and components of time series,
2. fitting and plotting of various growth curves and trend by Moving Average method,
3. Applications to real data by means of laboratory assignments.
4. Interpret and use a range of index numbers commonly used in the business sector.
5. Understand forecasting by different methods and calculate the variance of a random component.

11. Unit wise detailed content

1. Measurement of trend by method of moving averages (odd and even period)
2. Measurement of trend by method of Least squares (linear and parabola)
3. Determination of seasonal indices by method simple averages
4. Determination of seasonal indices by method of Ratio to moving averages
5. Determination of seasonal indices by method of Ratio to trend
6. Fitting of trend by Moving Average Method.
7. Computation of simple index numbers.
8. Computation of all weighted index numbers.



9. Computation of reversal tests. 10. Plotting ACF of a given time series. 11. Using Yule-Walker equation and Least squares to fit AR (1) and AR (2) models to real life data 12. Forecasting by exponential smoothing.
12. Web Links:
1. Christian Toninski, Heidrun Schumann, Interactive Visual Data Analysis, CRC press publication,2020 2. https://www.tutorialspoint.com/python_data_science/python_time_series.htm
13. Books Recommended
1. Christian Toninski, Heidrun Schumann, Interactive Visual Data Analysis, CRC press publication,2020 2. Aileen Nielsen, Practical Time Series Analysis, O'Reilly Media, Inc. 2019.


1. Name of the Department: Mathematics						
2. Course Name	Inferential Statistics	**L**	**T**		**P**	
3. Course Code	17200304	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 60		**Tutorials = 0**	**Practical = 0**			

8. Course Description:

This course introduces students to the basic theory behind the development and assessment of statistical analysis techniques in the areas of point and interval estimation, as well as hypothesis testing.

Topic includes: Point estimation and interval methods, including method of moments and maximum likelihood, unbiasedness, consistency, efficiency and sufficiency, hypothesis testing methods and related confidence interval.

9. Course Objectives:

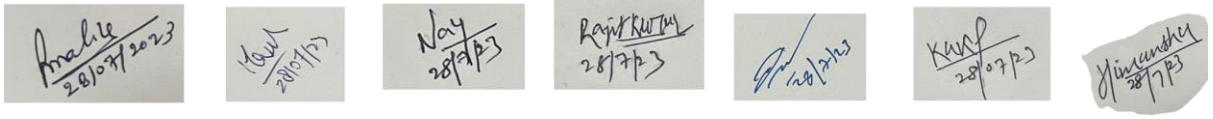
The objectives of the course are to:

1. Familiar the students about method of maximum likelihood and the properties of good estimators.
2. Familiar the students with the concept of statistical inference, point and interval estimation, hypothesis testing under a large variety of discrete and continuous probability models.
3. Familiar the students about ANOVA

10. Course Outcomes (COs):

Upon successful completion of this course the students are able to:

1. How to apply discrete and continuous probability distributions to various business problems.
2. Perform Test of Hypothesis as well as calculate confidence interval for a population parameter for single sample and two sample cases.
3. Learn non-parametric test such as the Chi-Square test for Independence as well as Goodness of Fit.



4. Show a working knowledge of sampling and sampling distributions.
5. Perform ANOVA and F-test

11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Estimator and their properties**

Parameter and statistic, sampling distribution and standard error of estimate. Point and interval estimation, Unbiasedness, Efficiency, Consistency and Sufficiency.

Unit - 2	**Number of lectures = 15**	**Title of the unit: Basic of Hypothesis and Method of Estimation**

Method of maximum likelihood estimation. Null and alternative hypotheses, Simple and composite hypotheses, Critical region, Level of significance, One tailed and two tailed tests, Types of errors, Neyman- Pearson Lemma.

Unit - 3	**Number of lectures = 15**	**Title of the unit: Large Sample Test**

Testing and interval estimation of a single mean, single proportion, difference between two means and two proportions.

Unit - 4	**Number of lectures = 15**	**Title of the unit: Small Sample Test and ANOVA Test**

Definition of Chi-square statistic, Chi-square tests for goodness of fit and independence of attributes. Definition of Student's't' and Snedcor's F-statistics. Testing for the mean and variance of univariate normal distributions, Testing of equality of two means and two variances of two univariate normal distributions. Related confidence intervals. Analysis of variance (ANOVA) for one-way and two-way classified data.

12. E-learning component

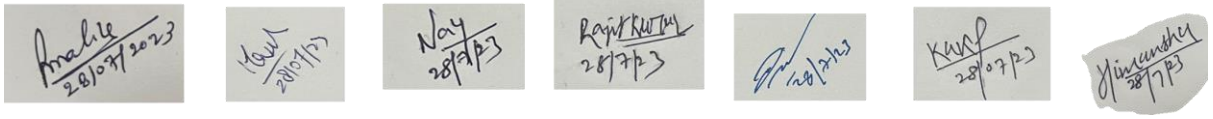
https://www.youtube.com/watch?v=VK-rnA3-41c

http://www.MathTutorDVD.com

https://www.youtube.com/watch?v=LlFHoQEq0Bg

13. Books Recommended

1. Robert V. Hogg, Joseph W. McKean and Allen T. Craig, Introduction to Mathematical Statistics, Pearson Education, Asia, 2007.
2. Gupta, S.C. and V.K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, New Delhi, 2008.



3. A.M. Mood, F.A. Graybill and D.C. Boes, Introduction to the theory of Statistics, McGraw Hill, 1974.
4. A.M. Goon, M.K. Gupta, and B. Das Gupta, Fundamentals of Statistics, Vol-II.
5. R.V. Hogg and A.T. Craig, Introduction to Mathematical Statistics.

1. Name of the Department: Mathematics						
2. Course Name	Operations Research	**L**	**T**		**P**	
3. Course Code	17200305	2	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd **(✓)**	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 30		**Tutorials = 0**		**Practical = 0**		

8. Course Description:

Operations Research is a science of modeling and optimization. It allows you to model real-world problems by using mathematics, statistics, and computers. It provides you tools and theories to solve these real-world problems by finding the optimal solutions to the model's subject to constraints of time, labor, resource, material, and business rules. With Operations Research, people make intelligent decisions to develop and manage their processes.

9. Course Objectives:

This module aims to introduce students to use quantitate methods and techniques for effective decision making; model formulation and applications that are used in solving decision making problems.

10. Course Outcomes (COs):

After completing this course, students will be able:

1. To formulate and solve mathematical models (linear programming problems) for a physical situation like production, distribution of goods and economics and solve them by graphical method.
2. To solve a LPP by simplex, Big-M and two-phase method.
3. To analyze the problem of transporting the products from origins to destinations with least transportation cost and solve it.
4. To understand the basic concept of Game theory and sequencing problem.
5. Use the acquired skillsets in developing innovative projects for the benefit of society.

11. Unit wise detailed content

Unit-1	**Number of lectures =8**	**Title of the unit: Linear Programming**

Operation research and its scope, Necessity of operations research in industry, Linear programming problems: Formulation and Graphical Solution, Convex sets, Convex linear combination, convex hull, hyper plane, theorems on convex sets.

Unit – 2	**Number of lectures = 7**	**Title of the unit: Simplex Method & Duality Theory**

Basic Solution, Degenerate Solution, Fundamental theorem of Linear Programming, Simplex method, Big-M and Two-Phase methods, Dual simplex method, Duality in linear programming, Weak Duality theorem, Basic Duality theorem, Fundamental theorem of Duality.

Unit – 3	**Number of lectures = 8**	**Title of the unit: Transportation and**

		Assignment Problem
Linear programming formulation of Transportation Problem, Initial basic feasible solution by North-west corner, Least-cost and Vogel's Approximation method, Test for optimality, Transportation Algorithm (MODI Method), Mathematical formulation of the Assignment Problem, Hungarian Assignment Method.		
Unit – 4	**Number of lectures = 7**	**Title of the unit: Game Theory**
Two-person zero- sum games, Pure strategies, Saddle point, Minimax-MaxiMin principle, Mixed Strategy Games, Solution of Mixed Strategy Games by Graphical and Linear programming Methods.		

12. E-learning component

https://www.youtube.com/watch?v=a2QgdDk4Xj
https://www.youtube.com/watch?v=ziHfSkOhAmg
https://www.youtube.com/watch?v=BUGIhEecipE
https://www.youtube.com/watch?v=RnZnIIksdwU
https://www.youtube.com/watch?v=h0bdo06qNVw
https://www.youtube.com/watch?v=533dp83Er6E

13. Books Recommended

1. Bazara, M. S., Jarvis, J. J., & Sherali, H. D. Linear programming and network flows, New Jersey: John Wiley & Sons, Inc, 2010 (4th edition).
2. Swarup, K., Gupta, P. K., Manmohan, Operations Research, Sultan Chand & Sons, 2010 (15th edition).
3. Taha, H. A., Operations research-an introduction, New Delhi: Pearson Prentice Hall (Indian print), 2017 (10th edition).
4. Hadley, G. Linear programming, New Delhi: Narosa Publishing House, 2002 (5th edition).
5. Anderson, D.R., Sweeney, D.J., Williams, T.A., Camm, J.D., and Martin, R.K. An Introduction to Management Science: Quantitative Approaches to Decision Making, Revised, Cengage Learning, 2011.

1. Name of the Department: Mathematics						
2. Course Name	Programming with Python	**L**	**T**		**P**	
3. Course Code	17200301	0	0		4	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AEC ()**	**SEC (✓)**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials =0**	**Practical = 60**

8. Course Description:

The goal of this course is to provide an introduction to Python. The course will discuss topics necessary for the participant to be able to create and execute Python programs.

9. Course Objectives:

1. Master the fundamentals of writing Python scripts.
2. Learn core Python scripting elements such as variables and flow control structures.
3. Discover how to work with lists and sequence data.
4. Write Python functions to facilitate code reuse.

10. Course Outcomes (COs):

By the end of this course, students should be able:

1. To install and run the Python interpreter.
2. To understand the concepts of file I/O and create and execute Python programs.
3. To enable to read data from a text file using Python.
4. To plot data using appropriate Python visualization libraries.
5. To plot data using appropriate Python visualization libraries.

11. Unit wise detailed content

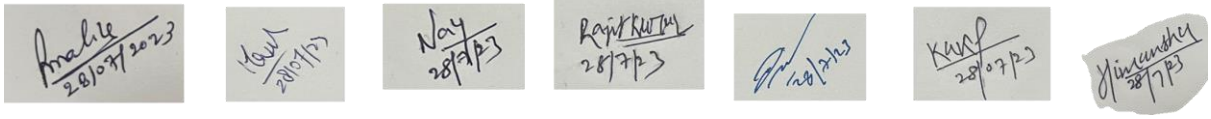
Unit-1	**Number of lectures = 30**	**Title of the unit: Introduction**

The Python Programming Language, History, features, Installing Python, Running Python program, Debugging, Syntax Errors, Runtime Errors, Semantic Errors, Experimental Debugging, Formal and Natural Languages, The Difference Between Brackets, Braces, and Parentheses.

Unit – 2	**Number of lectures = 30**	**Title of the unit: Variables and Expressions**

Values and Types, Variables, Variable Names and Keywords, Type conversion, Operators and Operands, Expressions, Interactive Mode and Script Mode, Order of Operations, Conditional Statements: if, if-else, nested if –else, Looping: for, while, nested loops, Control statements: Terminating loops, skipping specific conditions.

12. Brief Description of self-learning / E-learning component

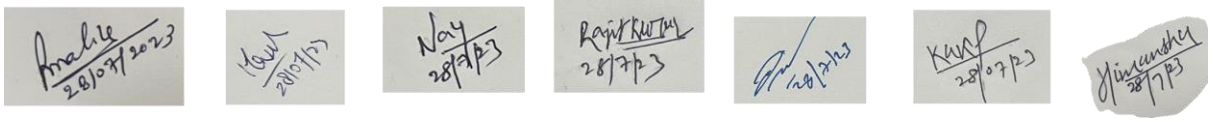


https://nptel.ac.in/courses/106106145

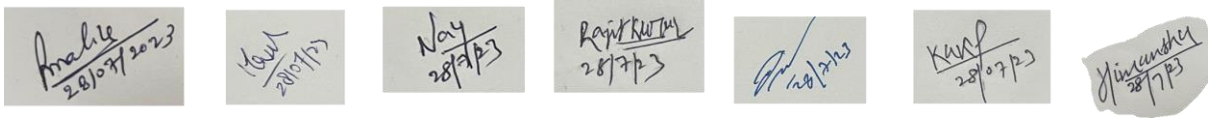
https://onlinecourses.nptel.ac.in/noc19_cs41/preview

13. Books Recommended

1. Jason Montojo, Jennifer Campbell, Paul Gries, An Introduction to Computer Science using Python 3, SPD, 2014.
2. P. K. Sinha & PritiSinha, “Computer Fundamentals”, BPB Publications, 2007.
3. T. Budd, Exploring Python, TMH, 1st Ed, 2011



Semester-IV



1. Name of the Department: Mathematics						
2. Course Name	Machine Learning-I	**L**	**T**		**P**	
3. Course Code	17200401	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

8. Course Description:

Machine Learning is the study of how to build computer systems that learn from experience. It is a subfield of Artificial Intelligence. This course is designed to develop the topics of support-vector machine, optimization, statistics and data sciences.

9. Course Objectives:

1. To introduce students to the basic concepts and techniques of Machine Learning.
2. To become familiar with regression methods, classification methods, optimization methods.
3. To understand a range of machine learning algorithms along with their strengths and weaknesses.
4. To be able to apply machine learning algorithms to solve problems of moderate complexity.

10. Course Outcomes (COs):

1. To identify machine learning techniques suitable for a given problem
2. To describe the problem of supervised learning from the point of view of function approximation, optimization, and statistics.
3. To implement mathematical concepts using real-world data.
4. To solve the problems using various machine learning techniques.
5. To identify machine learning techniques suitable for a given problem.

11. Unit wise detailed content

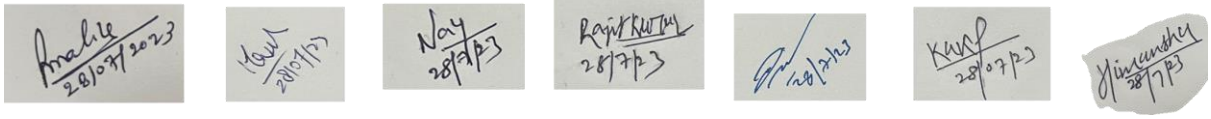
Unit-1	**Number of lectures = 20**	**Title of the unit: Overview of Machine Learning**

Introduction to Machine Learning, Types of learning approaches, steps in machine learning, machine representations of numbers, supervised and unsupervised learning, overflow, underflow, Basic operations of linear algebra, Norms, Matrix Decompositions.

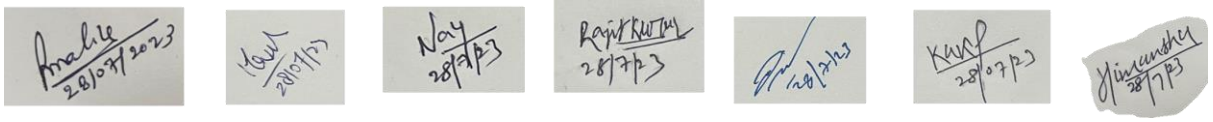
Unit – 2	**Number of lectures = 10**	**Title of the unit: Optimization**

Introduction to Constrained Optimization, Introduction to unconstrained Optimization, introduction to Numerical Optimization Gradient Descent-1.

Unit – 3	**Number of lectures = 15**	**Title of the unit: The Learning Paradigm**



Linear Regression, Least Squares Gradient Descent, Goodness of Fit, Bias-Variance Trade Off, Gradient Descent Algorithm, Variants of Gradient Descent, Maximum Likelihood Estimation (MLE).		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Classical Techniques**
Binary Trees, Bagging, Random Forests, support-vector machines (SVM), Naïve Bayes, k-Means, k-nearest neighbours (kNN), Gradient boosting.		
12. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/106106139 https://www.digimat.in/nptel/courses/video/106105152/L01.html https://nptel.ac.in/courses/106106213		
13. Books Recommended		
1. M P Deisenroth, A A Faisaland C S Ong, Mathematics for Machine Learning, Cambridge University Press, 2002. 2. Chrisopher Bishap, Pattern Recognition and Machine Learning, Springer, 2006. 3. Marco Gori, Machine Learning: A Constraint-Based Approach, Morgan Kaufmann. 2017 4. Ethem Alpaydin, Machine Learning: The New AI, MIT Press-2016. 5. Aurélien Géron, Hands-on ML with Scikit-Learn, Keras & TensorFlow, Second Edition, O'Reilly Media, Inc.		



1. Name of the Department: Mathematics						
2. Course Name	Machine Learning -1 Lab	**L**	**T**		**P**	
3. Course Code	17200402	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials = 0**	**Practical = 60**

8. Course Description:

This course is designed to emphasize the knowledge of Machine learning. Upon completion, students should be able to solve machine learning algorithm in R, Python and other software.

9. Course Objectives:

The objective of the course is

1. To understand the basic theory underlying machine learning.
2. To understand a range of machine learning algorithms along with their strengths and weaknesses.
3. To understand complexity of Machine Learning algorithms and their limitations.
4. To be capable of performing experiments in Machine Learning using real-world data.

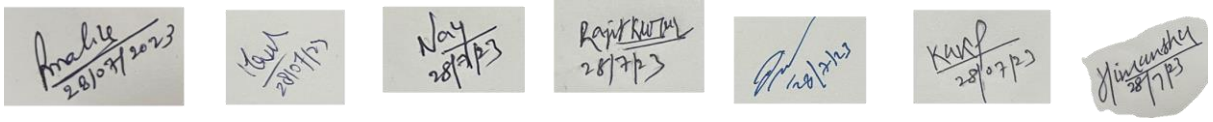
10. Course Outcomes (COs):

The students should be able to:

1. Understand the implementation procedures for the machine learning algorithms.
2. Design R/Python programs for various Learning algorithms.
3. Learn algorithmic topics of machine learning and mathematically deep enough to introduce the required theory.
4. Identify and apply Machine Learning algorithms to solve real world problems.
5. Demonstrate the Machine learning concepts for sustainable development.

11. The list of practical's to perform in the computer lab.

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
3. Write a program to demonstrate the working of the decision tree based algorithm.



6. Calculate the accuracy, precision, and recall for your data set.
7. Build an Neural Network by implementing the Back propag your data set's accuracy, precision, and recall ation algorithm and test the same using appropriate data sets.
8. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file.
6. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task.
7. Write a program to construct aBayesian network considering medical data.
8. Write a program to implement support vector machine.
9. Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set.
10. Write a program to implement Gradient Descent Algorithm.

12. E-learning component

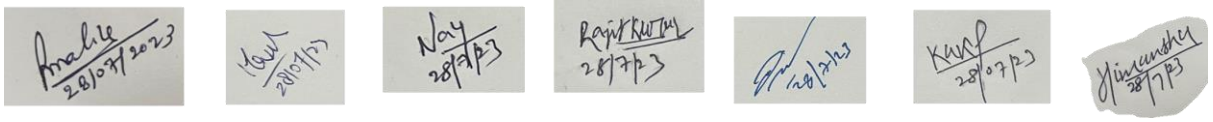
1. https://www.geeksforgeeks.org/k-nearest-neighbor-algorithm-in-python/
2. https://realpython.com/knn-python/
3. **https://nptel.ac.in/courses/106106213**

13. Books Recommended

1. Marco Gori, Machine Learning: A Constraint-Based Approach, Morgan Kaufmann. 2017

2. Ethem Alpaydin, Machine Learning: The New AI, MIT Press-2016.

3. Aurélien Géron, Hands-on ML with Scikit-Learn, Keras & TensorFlow, Second Edition, O'Reilly Media, Inc.



1. Name of the Department: Mathematics						
2. Course Name	Mathematical Foundations of Data Science	**L**	**T**		**P**	
3. Course Code	17200403	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even **()**	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 60	**Tutorials = 0**	**Practical = 0**

8. Course Description:

The focus is on the foundational computational statistical analysis and visualization methods underpinning modern data science. Currently available frameworks in the field like Regression analysis, Gradient Descent and Dimensionality Reduction, etc would be discussed.

9. Course Objectives:

1. Formulate simple problems, and code a high-level appropriate programme for data science.
2. Acquire knowledge of standard data visualization and formal inference procedures to interpret the results.
3. To develop complex statistical models to assess data and apply to real-world contexts.

10. Course Outcomes (COs):

By the end of this course, students should be able:

1. To develop relevant programming techniques of moderate complexity and execute in data science.
2. To explain the concept of real-life solutions.
3. To discuss the basic concepts of data science.
4. To develop the ability to build and assess data-based models.
5. To Demonstrate skills in writing mathematics.

11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Probability Review**

Random experiment, Sample space, Conditional probability, Density function, expected value, variance, joint, Marginal distribution, Bayes's rule.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Convergence and Sampling**

Sampling and Estimation, Probably Approximately Correct (PAC), Concentration of Measure, Markov Inequality, Chebyshev Inequality, Chernoff-Hoeffding Inequality, Union Bound and Examples, Importance Sampling, Sampling Without Replacement with Priority Sampling.

Unit – 3	**Number of lectures = 15**	**Title of the unit: Linear Regression**

Simple Linear Regression, Linear Regression with Multiple Explanatory Variables, Polynomial Regression, Cross-Validation, Other ways to Evaluate Linear Regression Models, Regularized Regression, Tikhonov Regularization for Ridge Regression, Lasso.

Unit – 4	**Number of lectures = 15**	**Title of the unit: Gradient Descent and Dimensionality Reduction**

Functions, Gradients, Gradient Descent, Learning Rate, Least Mean Squares Updates for Regression, Sum of Squared Errors Goal, Fitting a Model to Data, Data Matrices, Singular Value Decomposition, Eigenvalues and Eigenvectors, The Power Method.

12. Brief Description of self-learning / E-learning component

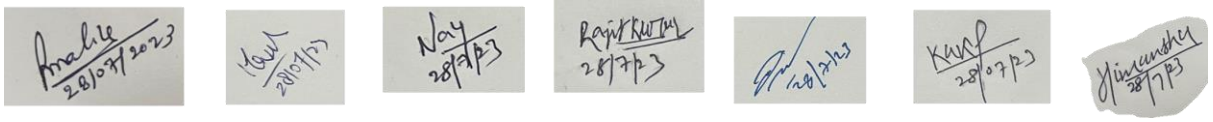
https://nptel.ac.in/courses/106106179

https://nptel.ac.in/courses/106106212

http://home.iitk.ac.in/~shalab/course5.htm

13. Books Recommended

1. Nina Zumal, John Mount (2014). Practical Data science in R, Managing Publication Company
2. Bernard Kolman, Robert C. Busby and Sharon Ross (2004). Discrete Mathematical Structures, New Delhi: Prentice Hall
3. V. Bhuvaneswari, T. Devi, (2016). Big Data Analytics: A Practitioner's Approach, Bharathiar University
4. 4. Jeffrey S. Saltz, Jeffre M. Stanton, "An Introduction to Data Science", Sage Publications.



1. Course Name	Statistical Quality Control and Demography	**L**	**T**		**P**	
2. Course Code	17200404	4	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

SQC refers to the application of statistical techniques to the monitoring and upkeep of product and service quality. When choosing whether to accept or reject a group of parts or items based on the quality of a sample, acceptance sampling is one technique that can be used.

8. Course Objectives:

This course will present the theory and methods of quality monitoring including process capability, control charts, acceptance sampling, quality engineering, and quality design.

9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To understand the basic concepts of quality monitoring.
2. To understand the statistical underpinnings of quality monitoring.
3. To learn various available statistical tools of quality monitoring.
4. To learn the statistical and economical design issues associated with the monitoring tools.
5. To demonstrate Statistical models and their daily life application.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit:** The Meaning of Quality and Quality Improvement

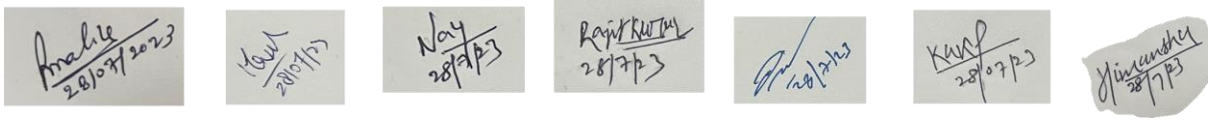
Brief History of Quality Methodology; Statistical Methods for Quality Control and Improvement; Total Quality Management: quality philosophy, links between quality and productivity, quality costs, legal aspects of quality implementing, quality improvement.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Methods of Statistical Process Control**

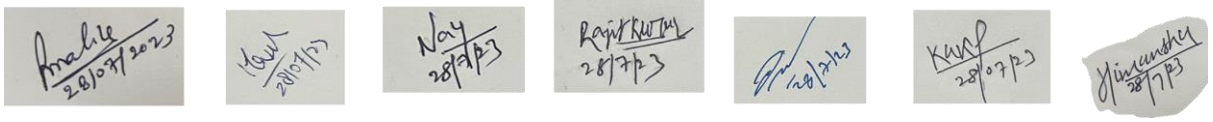
Chance and assignable causes, Statistical Basis of the Control Charts (basic principles, choices of control limits, significance of control limits, sample size and sampling frequency, rational subgroups, analysis of pattern on control charts, warning limits, Average Run Length-ARL)

Unit – 3	**Number of lectures = 15**	**Title of the unit: Statistical Control Charts**

Construction and Statistical basis of 3-s Control charts, Rational Sub-grouping, Control charts for variables: X-bar & R-chart, X-bar & s-chart. Control charts for attributes: np chart, p-chart.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Sampling Plans**
Single and double sampling plans, OC and ASN functions, Double and single Sampling plans for attributes using Binomial.		
11. Brief Description of self-learning / E-learning component		
https://www.youtube.com/watch?v=TbPUiJKyxqw https://www.youtube.com/watch?v=qb3mvJ1gb9g https://nptel.ac.in/courses/116102019		
12. Books Recommended		
1. Montogomery, D. C. (2009): Introduction to Statistical Quality Control, 6th Edition, Wiley India Pvt. Ltd. 2. Hoyle, David (1995): ISO Quality Systems Handbook, Heinemann Publication. 2nd Edition, Butterworth 3. Goon A.M., Gupta M.K. and Dasgupta B. (2002): Fundamentals of Statistics, Vol. II, 8th Edn. The World Press, Kolkata. 4. Grant, E, L. and Laven Worth, R.S.: Statistical Quality Control, McGraw Hill.		



Semester-V



1. Name of the Department: Mathematics						
2. Course Name	Research Methodology	**L**	**T**		**P**	
3. Course Code	17200501	2	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AECC (✓)**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

8. Course Description:

This course will provide an opportunity for students to establish or advance their understanding of research through critical exploration of research language, ethics, and approaches. The course introduces the language of research, ethical principles and challenges, and the elements of the research process within quantitative, qualitative, and mixed methods approaches.

9. Course Objectives:

To enable researchers (M.Sc. students), irrespective of their discipline, in developing the most appropriate methodology for their research studies. To make them familiar with the art of using different research methods and techniques.

10. Course Outcomes (COs):

By the end of this course, students should be able:

1. To understand the literature of research, process of the research and identification of appropriate research topic.
2. To understand the authentication of the statistical data and interpretation of results of research topics.
3. To understand the skills and ethics of research articles for the suitable publications.
4. To formulate of the research problem, analyses and simulation.
5. To understand research design models.

11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Fundamental Research**

Research, types of research, Research vs research methods, Research process, Relevant and quality research. Problem-solving in engineering, Identifying research topic, Problem definition, literature survey, literature review, Literature review, Research Design.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Formulation and Simulation**

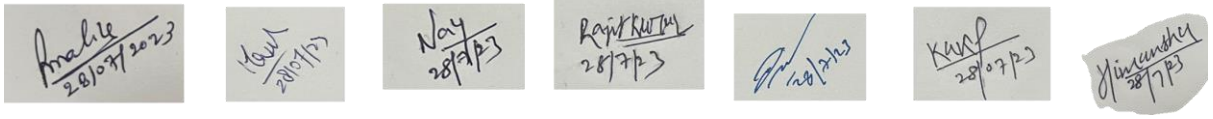
Models in general, Mathematical models, Model classifications, Modeling of engineering systems Theoretical models, Empirical models, Model evaluation, Limitations of mathematical models. Simulation models, Steps in a simulation study, Simulation software, Validation and data collection, Applications.

12. Brief Description of self-learning / E-learning component

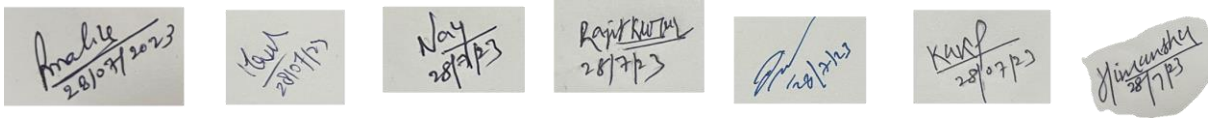
https://onlinecourses.nptel.ac.in/noc22_ge08/preview

https://archive.nptel.ac.in/courses/127/106/127106227/

13. Books Recommended



1. Research Methodology for Engineers- R Ganeshan, MJP Publishers. 2011.
2. Research Methodology- C R Kothari, New Age International, 2004.
3. Research Methodology: A step-by-step guide for beginners- Ranjit Kumar, Sage Publications, 2010.
4. Research Methods- R. Panneerselvan, Prentice Hall, 2004.



1. Name of the Department: Mathematics						
2. Course Name	Machine Learning-II	**L**	**T**		**P**	
3. Course Code	17200502	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

8. Course Description:

Machine Learning is the study of how to build computer systems that learn from experience. It is a subfield of Artificial Intelligence. This course is designed to develop the advanced topics of Recurrent Neural network, Decision Trees and Ensemble methods, natural language processing.

9. Course Objectives:

1. To introduce students to the basic concepts and advanced techniques of Machine Learning.
2. To become familiar with Neural network.
3. To understand a range of machine learning algorithms along with their strengths and weaknesses.
4. To be able to apply machine learning algorithms to solve problems of moderate complexity.

10. Course Outcomes (COs):

1. To identify machine learning techniques suitable for a given problem
2. To describe the problem of supervised learning from the point of view of function approximation, optimization, and statistics.
3. To implement mathematical concepts using real-world data.
4. To solve the problems using various advanced machine learning techniques.
5. To demonstrate machine learning models and their daily life application.

11. Unit-wise detailed content

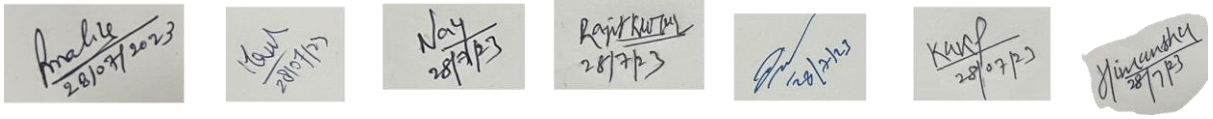
Unit-1	**Number of lectures = 15**	**Title of the unit:** Neural Networks

Multilayer Perceptron, Back propagation and their Applications, Convolutional, CNN Operations, CNN architectures, Training, Transfer Learning and their Application.

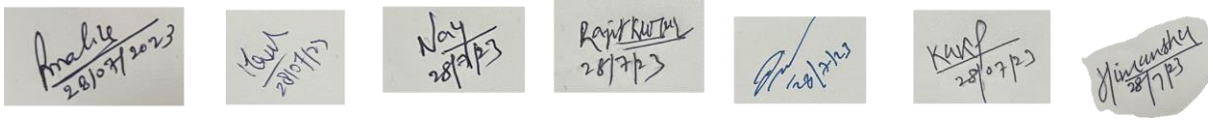
Unit – 2	**Number of lectures = 15**	**Title of the unit:** Recurrent Neural network

Activation Functions, Learning Rate decay, Weight initialization, Data Normalization, Batch Norm, Introduction to RNNs, Example - Sequence Classification, Training RNNs - Loss and BPTT, Vanishing Gradients and TBPTT, RNN Architectures, Deep RNNs and Bi- RNNs.

Unit – 3	**Number of lectures = 15**	**Title of the unit:** Decision Trees and Ensemble methods



Predictor form, Training Decision trees, Decision tree, classifiers, Learning Decision trees, Decision stumps, Stacking, Bagging and Boosting.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: N**atural Language Processing
Recurrent neural networks, back propagation through time, long short term memory, attention networks, memory networks, neural Turing machines, machine translation, speech recognition, syntactic and semantic parsing.		
12. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/106106139 https://www.digimat.in/nptel/courses/video/106105152/L01.html https://nptel.ac.in/courses/106106213		
13. Books Recommended		
1. M P Deisenroth, A A Faisaland C. S. Ong, Mathematics for Machine Learning, Cambridge University Press, 2002. 2. Chrisopher Bishap, Pattern Recognition and Machine Learning, Springer, 2006. 3. Marco Gori, Machine Learning: A Constraint-Based Approach, Morgan Kaufmann. 2017 4. Ethem Alpaydin, Machine Learning: The New AI, MIT Press-2016. 5. Aurélien Géron, Hands-on ML with Scikit-Learn, Keras & TensorFlow, Second Edition, O'Reilly Media, Inc.		



1. Name of the Department: Mathematics						
2. Course Name	Machine Learning –II Lab	**L**	**T**		**P**	
3. Course Code	17200503	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials = 0**	**Practical = 60**

8. Course Description:

This course is designed to emphasize the knowledge of Machine learning. Upon completion, students should be able to solve machine learning algorithm in R, Python and other software.

9. Course Objectives:

The objective of the course are:

1. To understand the basic theory underlying machine learning.
2. To understand a range of machine learning algorithms and their strengths and weaknesses.
3. To understand complexity of Machine Learning algorithms and their limitations.
4. To be capable of performing experiments in Machine Learning using real-world data.

10. Course Outcomes (COs):

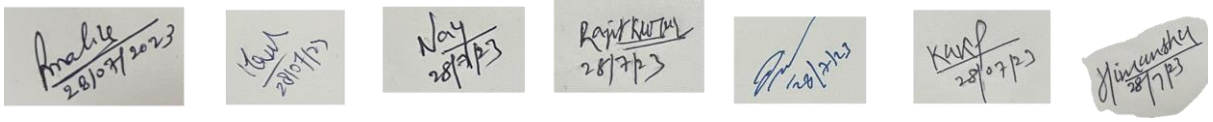
. The students should be able to:

1. Understand the implementation procedures for the machine learning algorithms.
2. Design R/Python programs for various Learning algorithms.
3. Learn algorithmic topics of machine learning and mathematically deep enough to introduce the required theory.
4. Identify and apply Machine Learning algorithms to solve real-world problems.
5. Demonstrate the concepts of Machine learning for sustainable development.

11. The list of practical to perform in the computer lab.

1. Importing datasets- Data visualization.
2. Extract the data from database using R/python.
3. Implement CNN Operations using python.
4. Implement an algorithm to demonstrate natural language processing.
5. Implement Deep RNNs using python.
6. Implement back propagation through time using R/Python.
7. Implement an algorithm to demonstrate the significance of genetic algorithm
8. Implement the finite words classification system using Back-propagation algorithm

9. Implement an algorithm to demonstrate Decision tree. 10. demonstrating Dimensionality Reduction
12. E-learning component
1. https://www.askpython.com/python/examples/backpropagation-in-python 2. https://realpython.com/knn-python/ 3. https://nptel.ac.in/courses/106106213
13. Books Recommended
1. Marco Gori, Machine Learning: A Constraint-Based Approach, Morgan Kaufmann. 2017 2. Ethem Alpaydin, Machine Learning: The New AI, MIT Press-2016. 3. Aurélien Géron, Hands-on ML with Scikit-Learn, Keras & TensorFlow, Second Edition, O'Reilly Media, Inc.



1. Course Name	Matrix Computations for Data Science	L	T		P	
2. Course Code	17200504	4	0		0	
3. Type of Course (use tick mark)		Core (✓)	DSE ()	AEC ()	SEC ()	OE ()
4. Pre-requisite (if any)		5. Frequency (use tick marks)	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	Tutorials = 0	Practical = 0

7. Course Description:

This course deals with applications of matrices to a wide range of areas of engineering and science. Initial some basics of linear algebra is discussed followed by matrix norms and sensitivity and condition number of the matrices.

8. Course Objectives:

The course focuses on iterative techniques for solving large sparse linear systems of equations which typically stem from the discretization of Singular value decomposition. In addition, computation of eigenvalues, least square problems and error analysis will be discussed.

9. Course Outcomes (COs):

Develop the skill set to

1. Explain and fluently apply fundamental linear algebraic concepts such as matrix norms, Eigen- and singular values and vectors
2. Estimate stability of the solutions to linear algebraic equations and eigenvalue problems.
3. Recognize matrices of important special classes, such as normal, unitary, Hermitian, positive definite and select efficient computational algorithms based on this classification.
4. Explain concepts of Eigenvalue Decomposition.
5. Demonstrate matrix and linear algebra models and their daily life application.

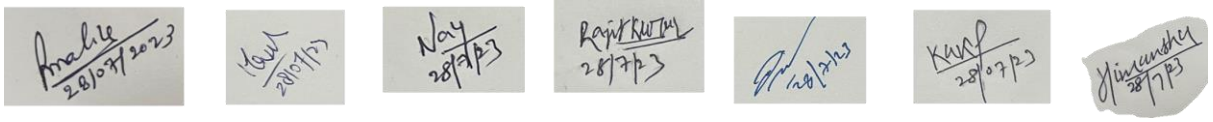
10. Unit wise detailed content

Unit – 1	Number of lectures = 15	Title of the unit: Introduction of Linear systems

Matrices and Gaussian Elimination, Matrix Notation, Matrix Multiplication Triangular Factors, Row Exchanges, Inverses and Transposes, Special Matrices and Applications All variants of Gaussian elimination and LU factorization, Cholesky factorization.

Unit – 2	Number of lectures = 15	Title of the unit: Linear least-squares problem

Normal equations, rotators and reflectors, QR factorization via rotators, reflectors and Gram Schmidt orthonormalization, QR method for linear least-squares problems, rank deficient least-squares problems.		
Unit – 3	**Number of lectures = 15**	**Title of the unit: Singular value decomposition**
Numerical rank determination via SVD, solution of least squares problems, Solution of linear least squares problems, normal equations, Rank deficient least squares problems, Sensitivity analysis of least-squares problems, Sensitivity of eigenvalues and eigenvectors.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Eigenvalue Decomposition**
Power, inverse power and Rayleigh quotient iterations, Schur's decomposition, unitary similarity transformation of Hermitian matrices to tridiagonal form, QR algorithm, implementation of explicit QR algorithm for Hermitian matrices.		
11. Brief Description of self-learning / E-learning component		
https://onlinecourses.nptel.ac.in/noc21_ma65/preview https://nptel.ac.in/courses/111102152 https://www.digimat.in/nptel/courses/video/111102152/L45.html		
12. Books Recommended		
1. L. N. Trefethen and David Bau, Numerical Linear Algebra, SIAM, Philadelphia, 1997. 2. D. S. Watkins, Fundamentals of Matrix Computation, 2nd Edition, Wiley, 2002. 3. L. Elden Matrix Methods in Data Mining and Pattern Recognition, SIAM, Philadelphia, 2007. 4. B.N. Datta, Numerical Linear Algebra and Applications.		



1. Name of the Department: Mathematics						
2. Course Name	Matrix Computations for Data Science Lab	**L**	**T**		**P**	
3. Course Code	17200505	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials =0**	**Practical = 60**

8. Course Description:

This course covers the concept of matrices, vector spaces, eigen values etc. It also covers Linear Transformations and algebra of linear transformations. This course also covers the concepts of Inner product spaces.

9. Course Objectives:

This course aims to provide a first approach to the subject of linear algebra, which is one of the basic pillars of modern mathematics. Linear Algebra course is to present the basic concepts of Vector Spaces and Linear Transformations. The course also presents basic concepts of Inner product Spaces.

10. Course Outcomes (COs):

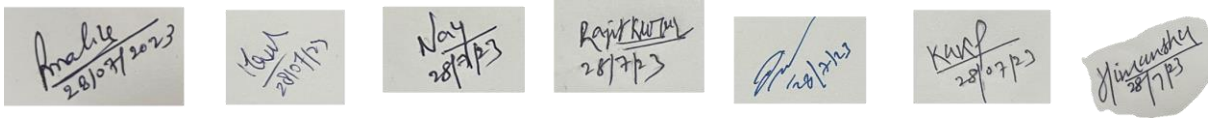
Upon completion of the course, students will be able to:

1. Demonstrate knowledge and understanding of the concept of Matrix.
2. Demonstrate ability to work within vector spaces with its properties.
3. Demonstrate ability to manipulate linear transformations.
4. Innovate new idea to implement for Research and development.
5. Understand the programming for better results.

11. Unit wise detailed content

Using Excel, MATLAB and Mathematica:

1. To find the determinants of matrices
2. To find an inverse of matrices
3. To find the rank of the matrices
4. To find the solution of system of linear and non-linear equations
5. To find the eigenvalues and eigenvectors matrices and linear transformations
6. To find the nullity of linear transformations



7. To find the roots of algebraic, Transcendental, Logarithmic equations
8. To find the solution of linear and non-linear differential equations
9. To find the solution of linear and non-linear partial differential equations
10. To solve the definite and indefinite integrals
11. To find the binary numbers
12. To draw the graph of polynomials in one and more than one variables

12. E-learning

1. https://in.mathworks.com/learn/tutorials/matlab-onramp.html
2. https://www.tutorialspoint.com/matlab/index.htm
3. https://in.mathworks.com/help/matlab/ref/eig.html

13. Books Recommended

1. Amos Gilat: MATLAB-An Introduction and its Applications, Wiley India Edition.
2. E. Balagurusamy: Programming in ANSI C, McGraw Hill Education, 8th Ed.

1. Name of the Department: Mathematics						
2. Course Name	Discrete Mathematics	**L**	**T**		**P**	
3. Course Code	17200506	4	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (Use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 60	**Tutorials = 0**	**Practical = 0**

8. Course Description:

Introduction to discrete structures and their applications like logic, gate and set theory, recursive programming, digital logic and combinatorial circuits, real number representation and finite automata used in computer science.

9. Course Objectives:

1. To provide basic and theoretical competencies that are majorly used in Computer Science.
2. To help students understand and appreciate the basic mathematical knowledge which is fundamental to Computer Science.
3. To understand the counting principles.
4. To discuss some important applications of Boolean algebra and graph theory in real life situations through switching circuits and shortest path algorithms.
5. Demonstrate the concepts of discrete mathematics sustainable development.

10. Course Outcomes (COs):

After the course, the student will be able to:

1. To determine the logical equivalence of propositions and the validity of formal arguments via truth tables.
2. Design and construction of a combinatorial circuit from a verbal description.
3. Understand switching circuits and their applications.
4. Understand Graphs and its applications in study of shortest path algorithms.
5. Demonstrate the concepts of discrete mathematics sustainable development

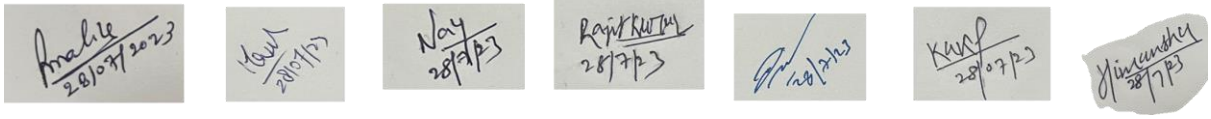
11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Sets, and Counting**

Sets, Relations and their properties, Equivalence relation, Partial order relation, Hasse diagram, Mathematical Induction, The basics of counting, Pigeon hole principle and its application, recurrence relations, solving recurrence relations, generating functions, inclusion-exclusion principle,

Unit – 2	**Number of lectures = 15**	**Title of the unit: Logical operators, Lattices**

Logic: Statement, Symbolic representation, Proposition, propositional logic, Compound proposition, logical operators, Tautologies and Contradiction, logical equivalence, validity of arguments, Quantifier, Predicates, Lattices, Lattices: Lattices as partially ordered sets and their properties.



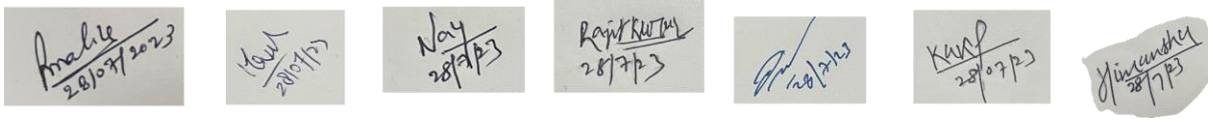
Unit – 3	Number of lectures = 15	Title of the unit: Boolean Algebra
Boolean Algebras, De Morgan's laws, Boolean homomorphism, Representation theorem; Boolean polynomials, Boolean Forms and their Equivalence. Minterm Boolean Forms, Sum of Products Canonical Forms. Disjunctive normal form and conjunctive normal form, Minimization of Boolean Functions, The Karnaugh Map method. Switching circuits and applications of switching circuits.		
Unit – 4	**Number of lectures = 15**	**Graph Theory**
Graph, Types of Graphs, Matrix Representations of Graphs, graph isomorphism, weighted undirected Graphs, Paths, cycles and circuits, connectivity, Subgraphs, Induced Subgraphs, Complete graphs, Bipartite graphs, Euler and Hamilton paths, Travelling salesman problem, shortest path, Dijkstra's algorithm, Euler's Theorem on the Existence of Eulerian paths and circuits. Trees, Spanning Trees. Minimal Spanning Trees and Kruskal's Algorithum, application of trees.		

12. Brief Description of self-learning / E-learning component

1. https://www.youtube.com/playlist?list=PL0862D1A947252D20
2. https://www.youtube.com/playlist?list=PLbMVogVj5nJQ-tIzyygzzLLxomzlbIiE4

13. Books Recommended

1. M. K. Venkataraman, N. Sridharan, N. Chandrasekaran: Discrete Mathematics, The National Publishing company, 2003.
2. C. L. Liu: Elements of Discrete Mathematics (2nd edition). McGraw Hill, 1985.
3. K. H. Rosen: Discrete Mathematics and Its Applications (8th edition), McGraw Hill, 2019.
4. Baburam: Discrete Mathematics, Pearson Education, 2012.



1. Course Name	Sampling Techniques	**L**	**T**		**P**	
2. Course Code	17200507	**4**	**0**		**0**	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical		
Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

Students majoring in math and statistics, as well as those majoring in the natural and social sciences who have already taken an introductory statistics course, will all benefit from this advanced statistics course. Students who have only taken our introductory statistics course (without a prior knowledge of probability) need to be comfortable with their statistical knowledge and be prepared to learn a new statistical computing programme. Students who want to major in mathematics and statistics must take this course, which will be offered every two to three years as one of our advanced "Topics in Statistics" courses.

8. Course Objectives:

Beginning with the simple random sample, stratification, and clustering, this course will concentrate on these fundamental concepts before moving on to more advanced designs like two-phase and multistage designs. The concept of sampling weights for a survey will be discussed, as well as inference and estimation techniques for the majority of these designs. Additionally, this course will cover how to graph complex survey data and how to use chi-square and regression tests to uncover relationships in complex survey data.

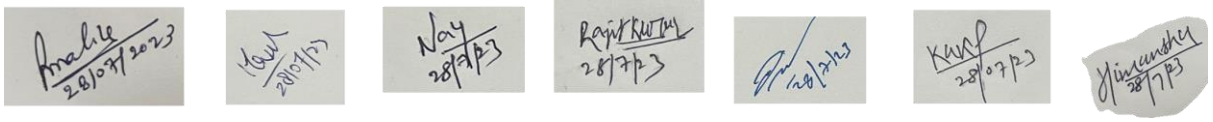
9. Course Outcomes (COs):

After completing the course, students are expected to be able:

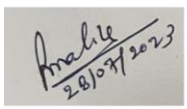
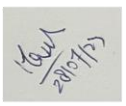
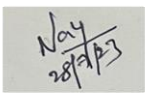
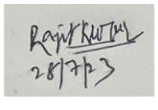
1. To understand the objectives of a sample survey.
2. know the "common" sampling designs, recognize the design in an application, and understand when it is appropriate to use each design.
3. To understand how we can compare sampling designs or estimators, and know what properties make a "better" design or estimator.
4. To analyze survey data using the statistical software R and understand the sampling methodology used in many research and government surveys.
5. To demonstrate Statistical models and their daily life application.

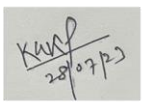
10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Introduction**



Sampling techniques, Sample surveys, type of surveys, Principal steps in a sample survey, Variability control in sample surveys, data collection, sampling unit, sampling frame, random sample, non-random sample.		
Unit – 2	**Number of lectures = 15**	**Title of the unit: Simple Random Sampling and Stratified sampling**
Simple random sampling without replacement (SRSWOR), Simple random sampling with replacement (SRSWR), Sampling For Proportions and Percentages, Stratified sampling, Advantages of stratified sampling.		
Unit – 3	**Number of lectures = 15**	**Title of the unit: Cluster Sampling and Systematic Sampling**
Elements, Clusters, area sampling, construction of clusters, Comparison between SRS and cluster sampling, linear systematic sampling, Systematic sampling in two dimensions.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Regression Method of Estimation**
Basics and fundamentals, Regression estimates with pre-assigned regression coefficient, estimate of variance, Regression estimator, Estimate of variance, Combined regression estimator, Regression method in stratified sampling.		
11. Brief Description of self-learning / E-learning component		
http://home.iitk.ac.in/~shalab/sampling/chapter1-sampling-introduction.pdf https://www.youtube.com/watch?v=hXsc5V1QaZ0 https://www.youtube.com/watch?v=UNDju3yphEE http://home.iitk.ac.in/~shalab/sampling/chapter11-sampling-systematic-sampling.pdf http://home.iitk.ac.in/~shalab/course432.htm		
12. Books Recommended		
1. P. S.R.S. Rao. Sampling methodologies and applications, Chapman and Hall/CRC 2000 2. Z. Govindrajalu. Elements of sampling theory and methods, Prentice Hall, 1999 3. P. Mukhopadhyaya. Sampling, Prentice Hall of India, 1998. 4. W.G. Cochran. Sampling techniques, Wiley 5. P.V. Sukhatme, B.V.Sukhatme, S. Sukhatme and C. Asok. Theory of sample surveys with applications: IASRI, Delhi, 1984 6. Singh, D., and Chaudhary, F. S., Theory and Analysis of Sample Survey Design, New Age International, 2018.		




1. Course Name	Design of Experiment	**L**	**T**		**P**	
2. Course Code	17200508	4	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This is a fundamental course in planning experiments and data analysis. The goal of the course is to teach students how to efficiently, and effectively plan, design, and carry out experiments as well as how to analyze the collected data and draw valid conclusions. Issues related to design and statistical analysis are discussed.

8. Course Objectives:

To introduce the fundamental tenets and procedures of statistical experiment design.

Statistical principles are used to determine the significances of the effects of various factors on a given response when there is uncertainty.

9. Course Outcomes (COs):

After completing the course, students are expected to be able to:

1. Describe the purpose of robust construction and how it is applied in experimental design
2. Formulate and validate the experimental designs in agricultural, medical, biomedical Projects
3. Avails them to fetch the background concepts of Model formulation and validation
4. Accomplish research-oriented concepts given for statistical techniques required for experimental designs.
5. Use the acquired skillsets in developing innovative projects to benefit society.

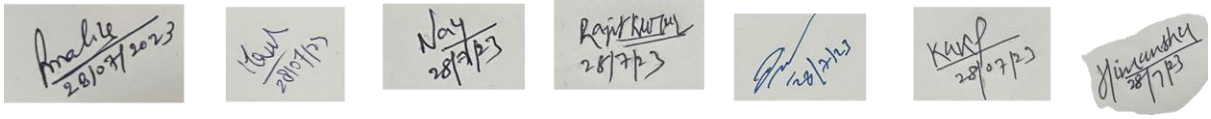
10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Experimental designs**

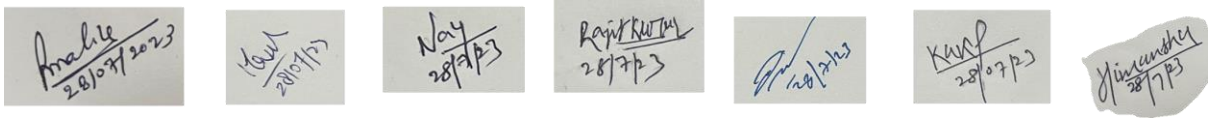
Strategy of Experimentation, Typical applications of Experimental design, Basic Principles, Guidelines for Designing Experiments, Introduction of analysis of variance (ANOVA)

Unit – 2	**Number of lectures = 15**	**Title of the unit: Basic Designs**

Completely Randomized Design (CRD), Randomized Block Design (RBD), Latin Square Design (LSD) – layout, model and statistical analysis, relative efficiency. Analysis with one missing observation in RBD and LSD.



Unit – 3	Number of lectures = 15	Title of the unit: Factorial Experiment
Factors, levels, interactions, treatment combination, randomization, Two-level experimental designs for two factors and three factors. Three-level experimental designs for two factors and three factors, Factor effects, Factor interactions, Fractional factorial design.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Randomized block design and Ranking Methods**
Split Plot Design in RBD and Strip arrangements, Groups of experiments with RBD and LSD, Measures of variability, Ranking method, Column effect method & Plotting method, Analysis of variance (ANOVA) in Factorial Experiment		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/110105087 https://www.youtube.com/watch?v=UNbG0j_Wz4k https://nptel.ac.in/courses/102106051		
12. Books Recommended		
1. Montgomery, D. C. (2008): Design and Analysis of Experiments, John Wiley. 2. Cochran, W.G. and Cox, G.M. (1959): Experimental Design. Asia Publishing House. 3. Das, M.N. and Giri, N.C. (1986): Design and Analysis of Experiments. Wiley Eastern Ltd. 4. Wu, C. F. J. And Hamada, M. (2009). Experiments, Analysis, and Parameter Design Optimization (Second edition), John Wiley.		



1. Course Name	Pattern Recognition	**L**	**T**		**P**	
2. Course Code	17200509	4	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

Automated systems that learn from experience can be designed using pattern recognition techniques. The methodologies, technologies, and algorithms of statistical pattern recognition are covered in this course from a number of angles.

8. Course Objectives:

To gain a thorough understanding of the theories, procedures, and applications of cutting-edge statistical pattern recognition and various mathematical techniques and their applications to the analysis and recognition of patterns in images and videos.

9. Course Outcomes (COs):

After completing the course, students are expected to be able to:

1. Idea about Pattern and Pattern Class, Design of a Pattern Recognition System
2. Idea of Instar, Outstar, Groups of Instar and Outstar, Different types of Memories.
3. Concept of Feedforward, Feedback and Competitive Learning Network
4. Concept of Gradient Descent Procedure and Newton's Algorithm.
5. Demonstrate Parametric and Non-Parametric statistical test models and their daily life application.

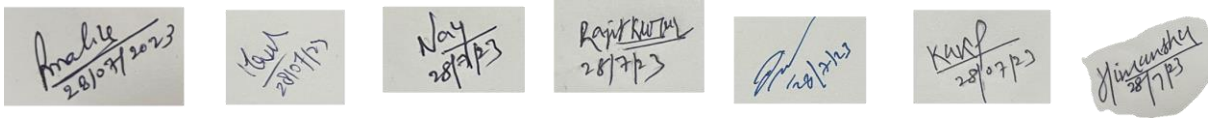
10. Unit-wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Pattern and Pattern Class**

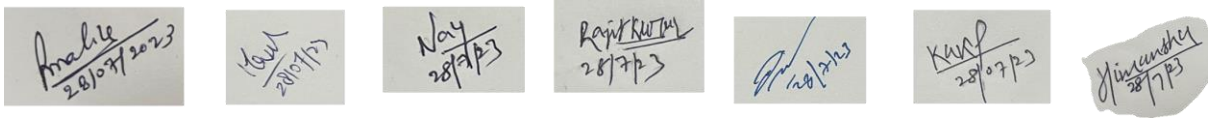
Design of a Pattern Recognition System, Syntactic and Decision Theoretic Approach, Bayesian Decision Theory, Continuous Features, Error, Risk and Loss.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Parametric and Non Parametric Methods**

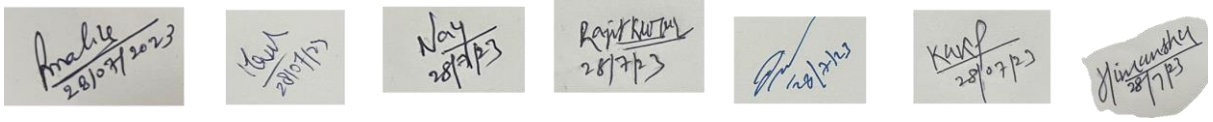
Histogram Method – Kernel Based Methods – K - Nearest Neighbour Method -- Probabilistic Neural Network base on Parzon Window.



Unit – 3	Number of lectures = 15	Title of the unit: Basics of ANN and pattern recognition task
Instar, Outstar, Groups of Instar and Outstar, Different types of Memories, Different PR Tasks by FF, FB and Competitive Learning Network, Pattern Clustering, Feature Mapping Problem, Different Feature Mapping Network, Self-Organizing Network.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: FF ANN and Single and Multilayer Network**
Pattern Association Network, Hebb's Law, Pattern Classification Network, Gradient Descent Procedure, Newton's Algorithm, Fixed Increment Learning, Variable Increment Learning, Support Vector Machine (SVM), Multilayer Neural Networks, and Unsupervised Learning.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/117105101 https://nptel.ac.in/courses/117106100 https://nptel.ac.in/courses/108108148		
12. Books Recommended		
1. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001 2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009 3. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006 4. I. H. Witten, Data Mining: Practical Machine Learning Tools And Techniques, 2nd Edn., Elsevier India, 2008		



Semester-VI



1. Name of the Department: Mathematics						
2. Course Name	Introduction to LATEX	**L**	**T**		**P**	
3. Course Code	17200601	0	0		2	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AEC ()**	**SEC (✓)**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials = 0**	**Practical = 60**			

8. Course Description:

This course introduces the basic concepts of LaTeX. Participants taking this course will be able to confidently create and design documents in LaTeX and presentations in Beamer.

9. Course Objectives:

Students will be able to understand.

1. To write mathematical formulae
2. To use tabular and array environments within a document.
3. Classifies non-linear programming problems.
4. To use Theorem, Corollary and other environments.

10. Course Outcomes (COs):

After completing this course students will be able:

1. To use the beamer package to create presentations.
2. To use BibTex to maintain bibliography information and to generate a bibliography for a particular document.
3. To analyze different types of documents.
4. To write complex mathematical formulae.
5. To understand the programming for better results.

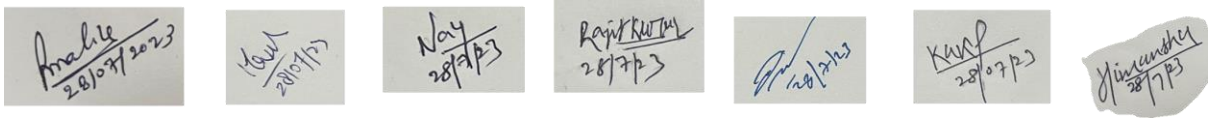
11. Unit wise detailed content

Unit-1	**Number of lectures = 30**	**Title of the unit: Introduction**

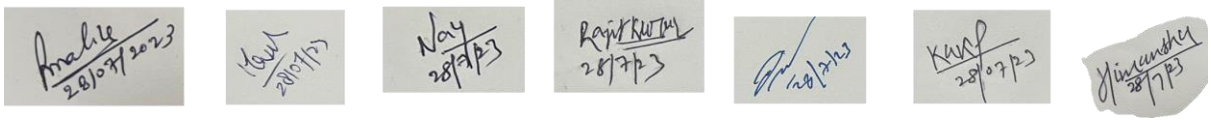
This topic introduces the learner to LaTeX, its installation, and different IDEs. The learner creates the first document using LaTeX, organizes content into sections using article and book class of LaTeX. In this topic, the session starts by reviewing different paper sizes, examines packages, formats the page by setting margins, customizing header and footer, changing the page orientation, dividing the document into multiple columns. The topic ends with reading different types of error messages.

Unit – 2	**Number of lectures = 30**	**Title of the unit: Formatting, insert tables and Images**

This topic concentrates on formatting text (styles, size, alignment), adding colors to text and entire page, and adding bullets and numbered items. It concludes by explaining the process of writing complex mathematics. The topic starts by creating basic tables, adding simple and dashed borders, merging rows and columns, and handling situations where a table exceeds



the size of a page. The sessions then continue to add an image, explore different properties like rotate, scale, etc.
12. Brief Description of self-learning / E-learning component
https://www.youtube.com/watch?v=0ivLZh9xK1Q&list=PL1D4EAB31D3EBC449 https://www.youtube.com/watch?v=bCumVPGR4ts&list=PL1D4EAB31D3EBC449&index=2 https://www.youtube.com/watch?v=kefvRACdXHs&list=PL1D4EAB31D3EBC449&index=3
13. Books Recommended
1. LaTex Tutorial: A primary by Indian Tex users' group 2. LaTeX for Complete Novices by Nicola L. C. Talbot 3. More Math into Latex 4th Edition by George Gratzer



1. Name of the Department: Mathematics						
2. Course Name	Applied Multivariate Statistical Analysis	**L**	**T**		**P**	
3. Course Code	17200602	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 60		**Tutorials = 0**		**Practical = 0**		

8. Course Description:

Multivariate statistical analysis provides ways to analyse dependence structures within multivariate data, as well as to meaningfully simplify, classify and group such data. The main theme of the course is the multivariate general linear model, and statistical methods include multivariate hypothesis testing, principal components, factor analysis, Classification, and clustering.

9. Course Objectives:

The objectives of this course are to:

1. To understand the fundamental concepts of Multivariate Data Analysis / Multivariate Statistical Analysis.
2. Conversant with various methods and techniques used in the summarization and analysis of multivariate data.
3. Develop feasible solutions to real-life problems, using principal component analysis, factor analysis and cluster analysis.
4. Conduct research using multivariate statistical models.

10. Course Outcomes (COs):

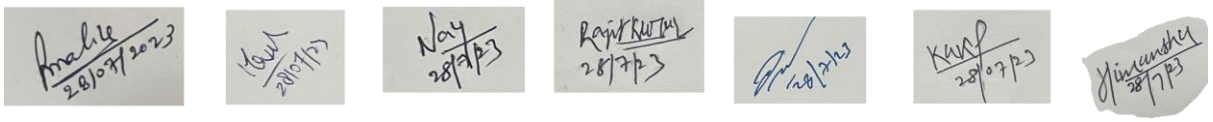
By the end of this course, students should be able:

1. To understand the basic concepts of Population & draw statistical inferences.
2. To select appropriate methods of multivariate data analysis.
3. To give knowledge about the basic features of factor analysis.
4. To introduce multivariate data & to find the principal components & Factors for further processing.
5. To use the acquired skillsets in developing innovative statistical projects for the benefit of society.

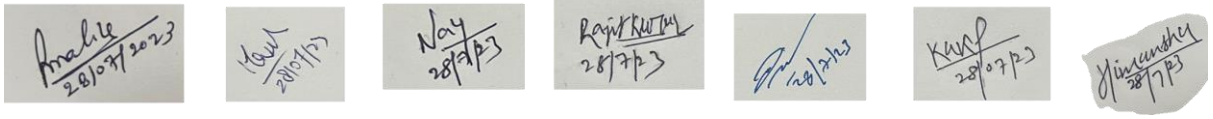
11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Introduction to Multivariate Data Analysis:**

Multivariate data and their diagrammatic representation. Exploratory multivariate data analysis, sample mean vector, sample dispersion matrix, sample correlation matrix, graphical representation, means, variances, co-variances, correlations of linear transforms.



Unit – 2	Number of lectures = 15	Title of the unit: Multivariate Linear Model
Introduction to multivariate normal distribution, probability density function and moment generating function of multivariate normal distribution, singular and non-singular normal distributions, Maximum likelihood estimation of parameters, tests of linear hypothesis, distribution of partial and multiple correlation coefficients and regression coefficients.		
Unit – 3	**Number of lectures = 15**	**Title of the unit: Logistic Regression**
Discriminant model and analysis, the decision process of discriminant analysis, Logistic Regression model and analysis, regression with a binary dependent variable, representation of the binary dependent variable, estimating the logistic regression model.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: multi-variate models**
Population and sample principal components, their uses and applications, large sample inferences, graphical representation of principal components, Biplots, the orthogonal factor model, dimension reduction, interpretation of factor analysis, Concepts of cluster analysis, similarity measures, Clustering based on statistical models.		
12. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/111104024 https://nptel.ac.in/courses/110105060 https://www.youtube.com/watch?v=hc7FYJIX6uU http://www.nitttrc.edu.in/nptel/courses/video/106106046/lec21.pdf		
13. Books Recommended:		
1. Anderson, T.W. (2003): An Introduction to Multivariate Statistical Analysis, 3rdEdn., John Wiley 2. Dean W. Wichern and Richard A. Johnson, Applied Multivariate Statistical Analysis, September 5, 2016 by Routledge published by Upper Saddle River, New Jersey, Pearson, 2019 3. William Mendenhall, Robert. J. Beaver, B. H. Beaver, Introduction to Probability & Statistics 12th Edition, Indian edition, Thomas. 4. Levin & Rubin, Statistics for Management (7th edition): Prentice Hall/Pearson Education 5. Gibbons, J. D. and Chakraborty, S (2003): Nonparametric Statistical Inference. 4th Edition. Marcel Dekker, CRC.		



1. Name of the Department: Mathematics						
2. Course Name	Applied Multivariate Statistical Analysis Lab	**L**	**T**		**P**	
3. Course Code	17200603	**0**	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials =0**	**Practical = 60**

8. Course Description:

The central theme of the course is the multivariate general linear model, and statistical methods include multivariate hypothesis testing, principal component analysis, factor analysis, discriminant analysis, canonical correlation analysis, and multivariate analysis of variance and covariance and cluster analysis

9. Course Objectives:

The objectives of this course are to:

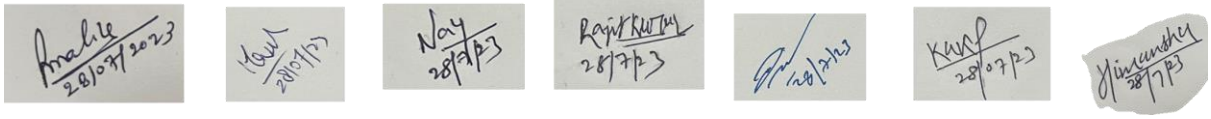
1. To understand the fundamental concepts of Multivariate Data Analysis Using R/Python.
2. Conversant with various methods and techniques using different software
3. Develop feasible solutions to real-life problems, using principal component analysis, factor analysis and cluster analysis.
4. Conduct research using multivariate statistical models.

10. Course Outcomes (COs):

1. Able to calculate the range of multivariate techniques.
2. Able to summarize and interpret multivariate data,
3. Have an understanding of the link between multivariate techniques and corresponding univariate techniques.
4. Use multivariate techniques appropriately.
5. Understand the programming for better results.

11. Detailed contents

1. Test for Multiple Correlation.
2. Test for sample Correlation matrix.
3. Perform means and variance using R/Python
4. Test probability density function
5. Multivariate Normal Distribution.
6. Perform discriminant models using different software.
7. Test for logistic regression
8. Principal Components Analysis.
9. Factor Analysis.



10. Test for clustering analysis.
11. Test for randomness based on total number of runs.
12. Kolmogorov -Smirnov test for one sample.

12. Brief Description of self-learning / E-learning component

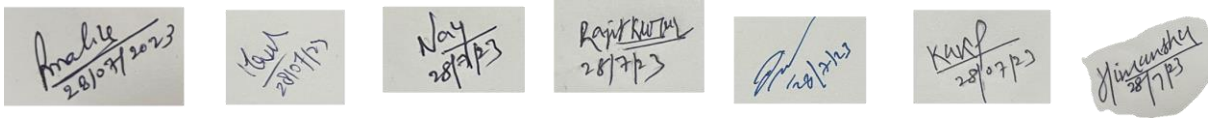
https://www.youtube.com/watch?v=z9XAXXGwUzM

https://www.geeksforgeeks.org/principal-component-analysis-with-python/

13. Books Recommended

1. Johnson, R.A., Wichern, D.W.: Applied Multivariate Statistical Analysis, Prentice-Hall, New Jersey, 2002.

2. Gibbons, J. D. and Chakraborty, S (2003): Nonparametric Statistical Inference. 4th Edition. Marcel Dekker, CRC.



1. Course Name	Reliability & Survival Analysis	**L**	**T**		**P**	
2. Course Code	17200604	4	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

The various modelling and evaluation techniques, also known as time-to-event data or survival data, are covered in this course. This course covers fundamental statistical ideas and techniques, such as log-rank and related tests, Kaplan-Meier graphs, survival and hazard functions, Cox proportional hazards model, and the extended Cox model for time-varying covariates. Participants in the course will also be expected to analyse survival analysis data using a practical statistical package (such as SAS, JMP, STATA, R, or S+).

8. Course Objectives:

In this course, you will learn how multiple linear regression models are created, the assumptions made by the models, how to determine whether the data supports the assumptions, and how to create and comprehend practical models.

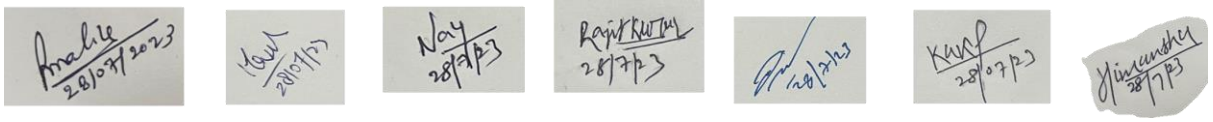
9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. To use probability and statistical theory to evaluate and identify appropriate methods of analysis.
2. To conduct appropriate statistical analysis of data to solve medical and public health problems.
3. To use a variety of statistical computer packages.
4. To apply probability and statistical methods to design experimental and observational studies in biomedical, clinical, and public health research.
5. To use the acquired skillsets in developing innovative projects for the benefit of society.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Survival function**



Survival function, Hazard function, cumulative hazard function, reversed hazard function, nature of hazard function, bath-tub shape hazard function, class of increasing failure rate distributions, decreasing failure rate distributions, theorems. Relations between survival function, probability function, hazard function, cumulative hazard function, reversed hazard function.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Lifetime distributions**

Survival distributions and their applications- exponential, gamma, Weibull, Rayleigh, lognormal distributions, and distribution having bath-tub shaped hazard function. Mean Residual Time.

Unit – 3	**Number of lectures = 15**	**Title of the unit: Censoring**

left censoring, right censoring, interval censoring, random censoring, times censoring, order censoring, hybrid censoring. Kaplan-Meier estimator of survival function, properties of Kaplan-Meier estimator, Nelson-Aalen estimator of cumulative hazards function. Linear and log-transformed confidence interval for survival function and cumulative hazard function. Q-Q plot, hazards plot for lifetime distributions.

Unit – 4	**Number of lectures = 15**	**Title of the unit: Reliability theory**

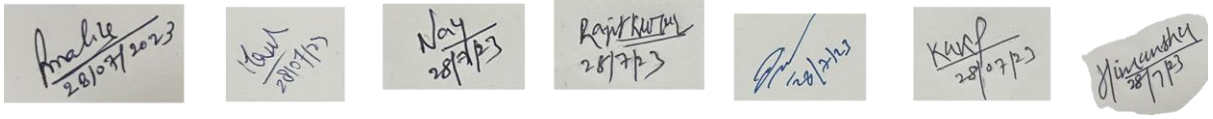
Reliability, systems with components in series, systems with parallel components, k-out-of-n systems, non-series parallel systems, systems with mixed mode failures, Standby redundancy: Simple standby system, k-out-of-n standby system.

11. Brief Description of self-learning / E-learning component

https://www.youtube.com/watch?v=fTX8GghbBPc
https://www.youtube.com/watch?v=Afpjct_ADC8
http://www.stat.columbia.edu/~madigan/W2025/notes/survival.pdf

12. Books Recommended

1. Lee, E.T. and Wang, J.W. (2003): Statistical Methods for Survival data Analysis, 3rd Edition, John Wiley and Sons.
2. Kleinbaum, D.G. (1996): Survival Analysis, Springer.
3. Chiang, C.L. (1968): Introduction to Stochastic Processes in Bio Statistics, John Wiley and Sons.
4. Indrayan, A. (2008): Medical Biostatistics, 2nd Edition Chapman and Hall/CRC.
5. Hosmer, D. and Lemeshow, S. (1999). Applied Survival Analysis: Regression Modeling of Time to Event Data, Wiley, New York.



1. Name of the Department: Mathematics						
2. Course Name	Reliability & Survival Analysis Lab	**L**	**T**		**P**	
3. Course Code	17200605	0	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 0	**Tutorials = 0**	**Practical = 60**

8. Course Description:

This course will teach you the various methods used for modeling and evaluating survival data or time-to event data.

9. Course Objectives:

The objective of the course is

1. Introduction to descriptive statistics
2. Introduction to statistical analysis
3. Examine survival function
4. Introduction to Lifetime distributions.
5. Understand R/ Python programming for better results.

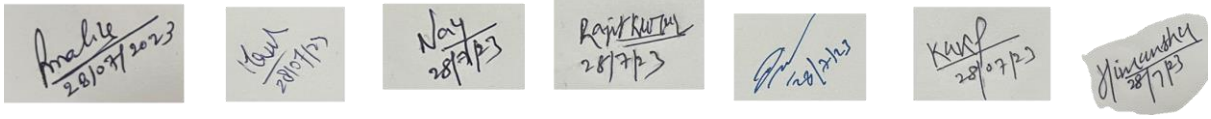
10. Course Outcomes (COs):

. The students should be able to:

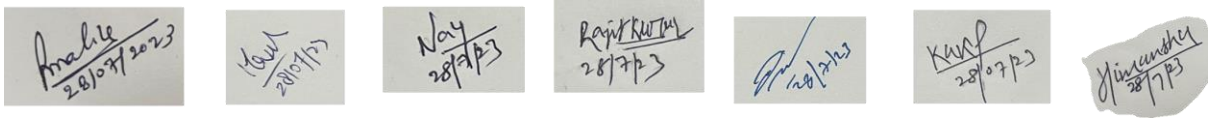
1. Understand the main features of SPSS
2. Use the SPSS GUI effectively
3. Perform **Lifetime distributions** with SPSS
4. Perform Survival distributions and their applications

11. The list of practical to perform in the computer lab.

1. Open, save, and close SPSS data and output files
2. Prepare a data entry codebook, Create a SPSS data file, Enter data into an SPSS data file.
3. Obtain descriptive statistics (Create a variety of graphs, histograms, bar graphs)



4. Manipulating the data to form 5. Kaplan-Meier using SPSS Statistics 6. Sorting the data file 7. Perform Survival distributions and their applications 8. Perform Weibull, Rayleigh and lognormal distributions 9. Draw Q-Q plot 10. Perform Regression models in Survival analysis
12. E-learning component
https://www.ibm.com/support/knowledgecenter/SSLVMB/welcome/ https://statistics.laerd.com/spss-tutorials/kaplan-meier-using-spss-statistics.php https://www.onlinespss.com/how-to-run-survival-analysis-test-in-spss/ https://www.youtube.com/watch?v=RvWxh4Bk0lc
13. Books Recommended
1. Field A., Discovering Statistics Using SPSS, Fourth Edition, SAGE, 2013. 2. Indrayan, A. (2008): Medical Biostatistics, 2nd Edition Chapman and Hall/CRC. 3. Hosmer, D. and Lemeshow, S. (1999). Applied Survival Analysis: Regression Modeling of Time to Event Data, Wiley, New York.



1. Course Name	Business Analytics	**L**	**T**		**P**	
2. Course Code	17200606	**2**	**0**		**0**	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

The business analytics course combines computer programming, business intelligence, and data analytics. The science of data analysis involves looking for patterns in data that can be used to create strategies. The use of it is widespread across almost all industries.

8. Course Objectives:

1. To make students understand the essentials of Business analytics.
2. To help students to analyze the steps involved in the Business Analytics process.
3. Enable students to Understand & Implement the Clustering algorithm.
4. To demonstrate the real time scenario by using Business Analytics Techniques

9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. Understand the essentials of Business analytics and the corresponding terminologies.
2. Analyze the steps involved in the Business Analytics process.
3. Illustrate competently on the topic of analytics.
4. Understand & Implement the Clustering algorithm.
5. Demonstrate the real time scenario (Case study) by using Business Analytics techniques.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 8**	**Title of the unit: Business Intelligence**

Introduction - History and Evolution: Effective and Timely decisions, Data Information and Knowledge, Architectural Representation, Role of Mathematical Models, Real-Time Business Intelligent System.

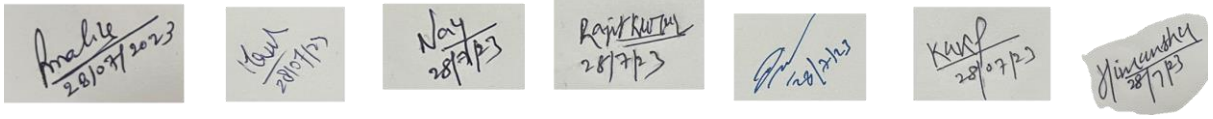
Unit – 2	**Number of lectures = 8**	**Title of the unit: Data Mining & Warehousing**

Data Mining - Introduction to Data Mining, Architecture of Data Mining and How Data mining works (Process) , Functionalities & Classifications of Data Mining, Representation of Input Data, Analysis Methodologies. Data Warehousing.

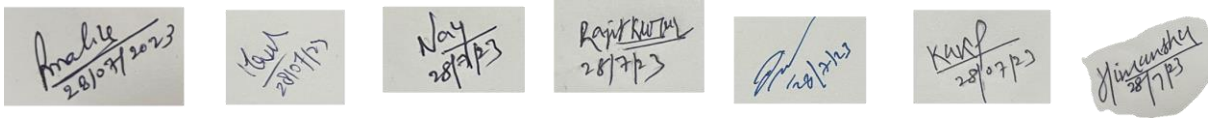
Unit – 3	**Number of lectures = 8**	**Title of the unit: Data Preparation**

Data Validation - Introduction to Data Validation, Data Transformation – Standardization and Feature Extraction.

Unit – 4	**Number of lectures = 6**	**Title of the unit: Data Analytics Process**



Analytics Process - Introduction to analytics process, Types of Analytical Techniques in BI – Descriptive, Predictive, Perspective, Social Media Analytics, Behavioral.
11. Brief Description of self-learning / E-learning component
https://onlinecourses.nptel.ac.in/noc20_mg11/preview https://archive.nptel.ac.in/courses/110/105/110105089/
12. Books Recommended
1. Larissa T.Moss & Shaku Atre,"Business Intelligence Roadmap : The Complete Project Lifecycle For Decision-Support Applications", First Edition, Addison-Wesley Professional, 2003 2. Kimball, R., Ross,M., Thornthwaite, W., Mundy, J., and Becker, B. John,"The Data Warehouse Lifecycle Toolkit: Practical Techniques for Building Data Warehouse and Business Intelligence". 3. Cindi Howson, "Successful Business Intelligence", Second Edition, McGraw-Hill Education, 2013.



1. Course Name	Data Visualization and Modelling	**L**	**T**		**P**	
2. Course Code	17200607	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

The goal of the Data Modeling and Visualization certification is to explore the field of data analysis using data modelling and visualisation techniques while minimising the risks and challenges related to data analytics. Through a virtual learning environment, the digital learning platform of TCS iON offers this online course. The course lasts for five to six months and includes online readings, lectures, digital tutorials, and other practical exercises and assignments.

8. Course Objectives:

1. To introduce students to the fundamental problems, concepts, and approaches in the design and analysis of data visualization systems.
2. To familiarize students with the stages of the visualization pipeline, including data modeling, mapping data attributes to graphical attributes, perceptual issues, existing visualization paradigms, techniques, and tools, and evaluating the effectiveness of visualizations for specific data, task, and user types.

9. Course Outcomes (COs):

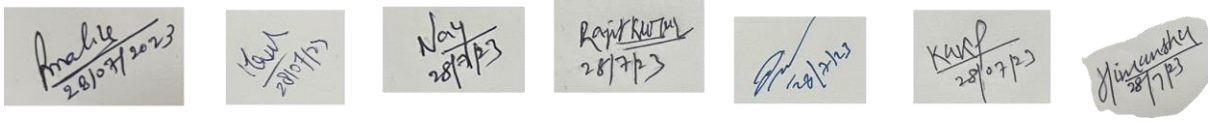
After completing the course, students are expected to be able:

1. To know the history of data visualization and its connection with computer graphics
2. To understand the visualization pipeline with its relationship to other data analysis pipelines.
3. To know the definition(s) of the visualization and interpretations of the notion.
4. To know categories of visualization and application areas.
5. To use the acquired skillsets in developing innovative projects for the benefit of society.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 06**	**Title of the unit: Visualizing Data and Coordinate Systems**

Mapping Data onto Aesthetics, Aesthetics and Types of Data, Scales Map Data Values onto Aesthetics, Cartesian Coordinates, Nonlinear Axes, Coordinate Systems with Curved Axes.



Unit – 2	**Number of lectures = 08**	**Title of the unit: Color Scales**
Color as a Tool to Distinguish, Color to Represent Data Values, Color as a Tool to Highlight, Directory of Visualizations and Visualizing Amounts: Amounts, Distributions, Proportions, x–y relationships, Geospatial Data, Bar Plots, Grouped and Stacked Bars, Dot Plots and Heatmaps.		
Unit – 3	**Number of lectures = 08**	**Title of the unit: Visualizing Distributions and Visualizing Proportions**
Empirical Cumulative Distribution Functions and Q-Q Plots, Empirical Cumulative Distribution Functions, Highly Skewed Distributions, Quantile-Quantile Plots, A Case for Pie Chart, A Case for Side-by-Side Bars, A Case for Stacked Bars and Stacked Densities, Visualizing Proportions Separately as Parts of the Total.		
Unit – 4	**Number of lectures = 08**	**Title of the unit: Visualizing Trends and Visualizing Geospatial**
Data Smoothing, Showing Trends with a Defined Functional Form, Detrending and Time-Series Decomposition, Projections, Layers, Choropleth Mapping, Cartograms.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/127101012 https://www.youtube.com/watch?v=JhK2qVi5dC4 https://www.coursera.org/specializations/data-analysis-visualization-foundations		
12. Books Recommended		
Danyel Fisher, Miriah Meyer, Making Data Visual: A Practical Guide to Using Visualization for Insight, O'Reilly Media, 2017. Claus O. Wilke, Fundamentals of Data Visualization, O'Reilly Media, Inc, 2019. Polley, David E. Börner, Katy, a practical guide to making sense of data, The MIT Press, 2014. Wickham, H. 2016. ggplot2: Elegant Graphics for Data Analysis. 2nd ed. New York: Springer.		

1. Course Name	Graph Theory	**L**	**T**		**P**	
2. Course Code	17200608	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This course covers basic concepts of Graph theory: meaning and different types of graphs. This course also covers the concepts of Trees, spanning tree, Circuits and Directed Graphs etc.

8. Course Objectives:

The objective of this course is to present the basic concepts of Graph Theory: Graphs, Types of graphs, Eulerian and Hamiltonian graphs, directed Graphs, Tress, Circuits, Planar Graphs, Graph Coloring etc

9. Course Outcomes (COs):

On completion of this course, students will be able to:

1. Understand the basic concepts of graph theory.
2. Illustrate the fundamental applications of Graph Theory in different walks of life.
3. Implement the graph theory in real-life problems that can be modelled into graphs.
4. Demonstrate ability to work with Planar graphs and Coloring of Graphs.
5. Use the acquired skillsets in developing innovative graph theory-based projects for the benefit of society.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 6**	**Title of the unit: Introduction to Graphs**

Definition of a graph, finite and infinite graphs, incidence of vertices and edges, types of graphs, subgraphs, walks, trail, paths, Circuits, Cycles, Sub graphs, Induced and Spanning subgraphs.

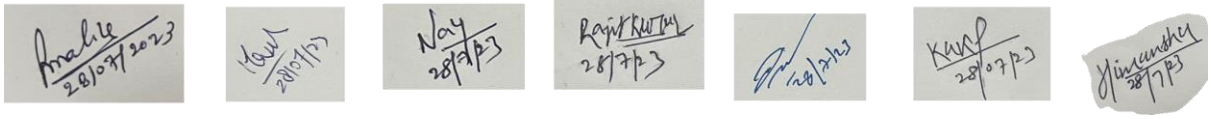
Unit – 2	**Number of lectures = 8**	**Title of the unit: Trees**

Trees and Fundamental Circuits: Definitions: Trees, Spanning trees, Some Properties of trees (no proof). Rooted and binary tree. Finding all the spanning trees of a graph and Spanning trees in a weighted graph. Traversal of Binary Tree, Pre-order and Post-order Traversal. Prefix codes, optimal tree.

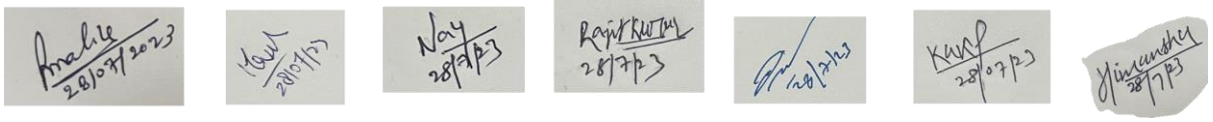
Unit – 3	**Number of lectures = 8**	**Title of the unit: Coloring and Dominating sets**

Definition of Chromatic number. Chromatic Partitioning. Chromatic Polynomial. Finding Chromatic polynomial by Decomposition Theorem and by Multiplication Theorem (without Proofs). Dominating set. Minimal Dominating set.

Unit – 4	**Number of lectures = 8**	**Title of the unit: Directed Graphs**



Directed Graphs: Types of digraphs, digraphs and binary relations, directed paths and connectedness, Euler digraphs.
11. Brief Description of self-learning / E-learning component
https://nptel.ac.in/courses/111106102 https://youtu.be/q0woiOp7sqU https://nptel.ac.in/courses/106108054 https://www.youtube.com/watch?v=tx9VbLwxl6k
12. Books Recommended
1. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Prentice – Hall of India Pvt. Ltd, 2004. 2. F. Harary: Graph Theory, Addition Wesley, 1969. 3. Seymour Lipschutz and Marc Lipson, Theory and Problems of Discrete Mathematics, Schaum Outline Series, McGraw-Hill Book Co, New York, 2007. 4. John A. Dossey, Otto, Spence and Vanden K. Eynden, Discrete Mathematics, Pearson, Fifth Edition, 2005.



1. Course Name	Operating System	**L**	**T**		**P**	
2. Course Code	17200609	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This course will provide an introduction to operating system design and implementation. The operating system provides a well-known, convenient, and efficient interface between user programs and the bare hardware of the computer on which they run. The operating system is responsible for allowing resources (e.g., disks, networks, and processors) to be shared, providing common services needed by many different programs.

8. Course Objectives:

1. To understand the basic concepts and functions of operating systems.
2. To understand Processes and Threads
3. To analyze Scheduling algorithms.
4. To understand the concept of Deadlocks.
5. To analyze various memory management schemes.

9. Course Outcomes (COs):

After completing the course, students are expected to be able to :

1. Describe the general architecture of computers.
2. Describe, contrast and compare differing structures for operating systems.
3. Understands the different services provided by Operating System at different level.
4. They learn real life applications of Operating System in every field.
5. Understands the use of different process scheduling algorithm and synchronization techniques to avoid deadlock.

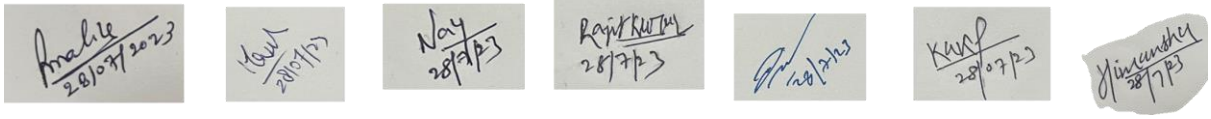
10. Unit wise detailed content

Unit – 1	**Number of lectures = 08**	**Title of the unit: Operating system overview:**

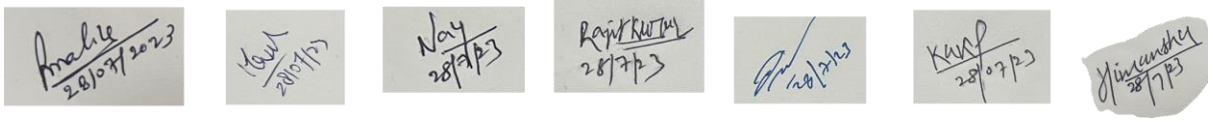
Computer System Overview-Basic Elements, Instruction Execution, Interrupts, Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. Operating system overview-objectives and functions, Evolution of Operating System

Unit – 2	**Number of lectures = 08**	**Title of the unit: Process management**

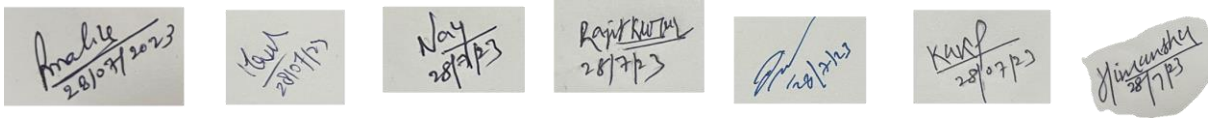
Processes — Process Concept, Process Scheduling, Operations on Processes, Inter-process Communication; CPU Scheduling — Scheduling criteria, Scheduling algorithms, Multiple-processor scheduling, Real time scheduling; Threads- Overview, Multithreading models.



Unit – 3	Number of lectures = 8	Title of the unit: Storage Management
Main Memory — Background, Swapping, Contiguous Memory Allocation, Paging, Segmentation, Segmentation with paging, 32 and 64-bit architecture Examples Virtual Memory — Background, Demand Paging, Page Replacement, Allocation.		
Unit – 4	**Number of lectures = 8**	**Title of the unit: File systems**
Mass Storage system — Overview of Mass Storage Structure, Disk Structure, Disk Scheduling and Management, swap space management; File-System Interface — File concept, Access methods, Directory Structure, Directory organization, File system mounting, File Sharing and Protection.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/106105214 https://nptel.ac.in/courses/106102132 https://www.youtube.com/watch?v=g7ROQmlY9vM		
12. Books Recommended		
1. Operating System Concepts, 9th edition Peter B. Galvin, Greg Gagne, Abraham Silberschatz, John Wiley & Sons, Inc. 2. Modern Operating Systems -By Andrew S. Tanenbaum (PHI) 3. Operating Systems 5th Edition, William Stallings, Pearson Education India 4. Sumitabha Das, Unix Concepts & Applications includes SCO UNIX & Linux, Tata McGraw Hill.		



Semester-VII



1. Name of the Department: Mathematics						
2. Course Name	Hadoop Programming	**L**	**T**		**P**	
3. Course Code	17200701	**0**	0		4	
4. Type of Course (use tick mark)		**Core ()**	**DSE ()**	**AEC ()**	**SEC (✓)**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials =0**	**Practical = 60**			

8. Course Description:

Hadoop Courses will teach you how to design distributed systems to manage big data using Hadoop and related technologies. This course will teach you how to manage Big Data on a cluster using HDFS and MapReduce, as well as how to design real-world systems with the Hadoop ecosystem.

9. Course Objectives:

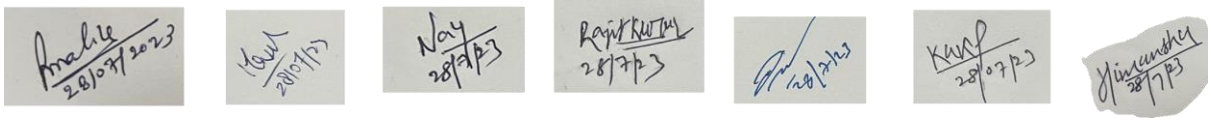
The Big Data and Hadoop training course is intended to provide students with the knowledge and skills necessary to become a successful Hadoop Developer. This training will provide in-depth knowledge of concepts such as Hadoop Distributed File System, Hadoop Cluster - Single and Multi-node, Hadoop 2.0, Map-Reduce, and others.

10. Course Outcomes (COs):

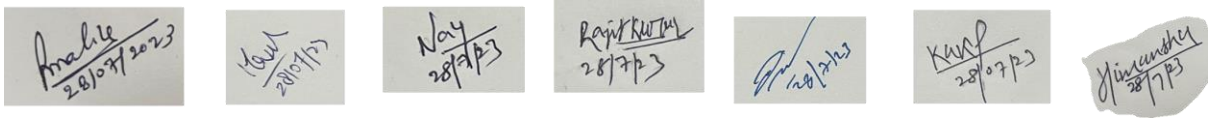
1. Understand Hadoop framework and Hadoop Distributed File system.
2. Illustrate the concepts of NoSQL using MongoDB and Cassandra for Big Data.
3. Demonstrate the Map Reduce programming model to process the big data along with Hadoop tools.
4. Apply machine Learning algorithms for real world big data., web contents and Social Networks to provide analytics with relevant visualization tools.
5. Demonstrate the concepts of Hadoop programming for sustainable development.

11. Detailed contents

1. Basic Concepts of Hadoop and its features
2. The Hadoop Distributed File System (HDFS)
3. Anatomy of a Hadoop Cluster: Hadoop cluster modes, Hadoop Architecture,
4. Hadoop Storage - Hadoop daemons (Name node-Secondary name node-Job tracker-Task tracker-Data node,etc)
5. Anatomy of Read & Write operations – Interacting HDFS using command-line (HDFS Shell and FS shell commands)
6. Implement Matrix Multiplication program with Hadoop Map Reduce.
7. Implement file management tasks in Hadoop.
8. Hadoop Cluster Setup
9. Write a MapReduce Application to implement Combiners.
10. Write a MapReduce Application to implement Reduce-side Join.



11. Write a MapReduce Application to implement Map-side Join
12. Brief Description of self-learning / E-learning component
https://www.tutorialspoint.com/hadoop/hadoop_hdfs_overview.htm https://freevideolectures.com/course/4233/nptel-big-data-computing/3 https://www.youtube.com/watch?v=0QIv4kk18Kg
13. Books Recommended
1. Boris lublinsky, Kevin t. Smith Alexey Yakubovich, “Professional Hadoop Solutions”. Wiley, 2015. 2. Chris Eaton, Dirk Deroos et al., “Understanding Big Data”, McGraw Hill, 2010. 3. Tom White, “HADOOP”: The definitive Guide”, O Reilly 2012. 4. Srinath Perera, Thilina Gunarathne, "Hadoop MapReduce Cookbook", PACKT publishing, 2013.



1. Name of the Department: Mathematics						
2. Course Name	Database Management System	**L**	**T**		**P**	
3. Course Code	17200702	4	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical =0**

8. Course Description:

The Database Management System course presents the fundamental concepts of database design and use. It provides a study of data models, data description languages, and SQL, transactions and their properties, Normalization, Relational Algebra and Calculus and object databases.

9. Course Objectives:

1. To understand the basics of data base system
2. To learn Query Basics and SQL commands
3. To comprehend the concepts of storage structures, form design, usage of report.

10. Course Outcomes (COs):

By the end of this course, students should be able to:

1. Create a normalized database for an organization.
2. Implement and test database queries for any real time data bases.
3. Formulate and design forms and reports for database applications.
4. Apply normalization for the development of application software's.
5. Use the acquired skillsets in developing innovative projects for the benefit of society.

11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit: Introduction to Databases and Transactions**

Advantages and Components of a Database Management Systems, Feasibility Study, Class Diagrams, Data Types, Events, Normal Forms, Integrity, Converting Class Diagrams to Normalized Tables, Data Dictionary.

Unit – 2	**Number of lectures = 15**	**Title of the unit: Data Models**

The importance of data models, Basic building blocks, Business rules, the evolution of data models, Degrees of data abstraction, Database Design, ER-Diagram and Unified Modeling Language, Database design and ER Model: overview, ER-Model, Constraints.

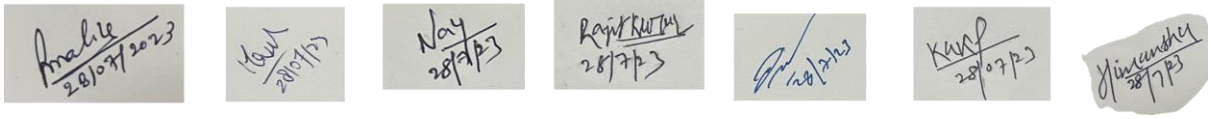
Unit – 3	**Number of lectures = 15**	**Title of the unit: Queries and subqueries**

Query Basics, Computation Using Queries, Subtotals and GROUP BY Command, Queries with Multiple Tables, Subqueries, Joins, DDL & DML, Testing Queries.

Unit – 4	**Number of lectures = 15**	**Title of the unit: Constraints and Views**

What is constraints, types of constraints, Integrity constraints, Views: Introduction to views, data independence, security, updates on views, comparison between tables and views.

12. Brief Description of self-learning / E-learning component



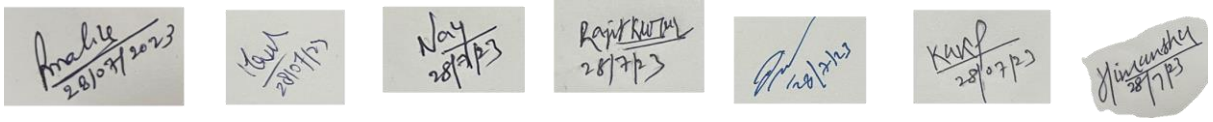
https://nptel.ac.in/courses/106105175

https://nptel.ac.in/courses/106106095

https://www.svecw.edu.in/Docs%5CITIIBTechIISemLecDBMS.pdf

13. Books Recommended

1. Fundamentals of Database Systems, Elmasri Navathe, Seventh edition (2021) Pearson Education.
2. An Introduction to Database systems, C.J. Date, A.Kannan, S.Swami Nadhan, Pearson, Eight Edition (2006).
3. Data base Management Systems, Raghurama Krishnan, Johannes Gehrke, TATA McGrawHill 3rd Edition (2014).
4. A. Silberschatz, H.F. Korth and S. Sudharshan, "Database System Concepts" , Fifth Edition, Tata McGraw Hill, New Delhi, 2006.



1. Name of the Department: Mathematics						
2. Course Name	Database Management system lab	**L**	**T**		**P**	
3. Course Code	17200703	**0**	0		4	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials =0**	**Practical = 60**			

8. Course Description:

This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business scientific and engineering applications at present.

9. Course Objectives:

1. To explain basic database concepts, applications, data models, schemas and instances.

2. To emphasize the importance of normalization in databases.

3. To facilitate students in Database design

4. To familiarize issues of concurrency control and transaction management.

10. Course Outcomes (COs):

At the end of the course the students are able to:

1. Apply the basic concepts of Database Systems and Applications.

2. Use the basics of SQL and construct queries using SQL in database creation and interaction.

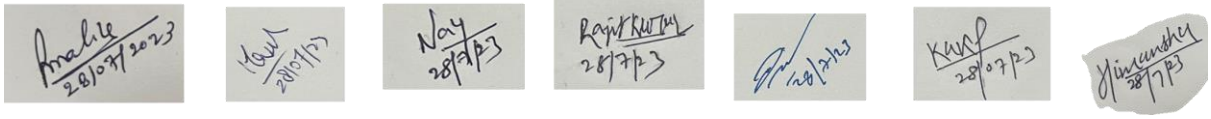
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.

4. Analyze and Select storage and recovery techniques of database system

5. Learn through experiments, innovate, new ideas to implement for Research and development.

11. Detailed contents

1. Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.).
2. Student should decide on a case study and formulate the problem statement.
3. To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form.
4. Overview of using SQL tool, Data types in SQL.
5. Creating Tables (along with Primary and Foreign keys).
6. Altering Tables and Dropping Tables using MySQL.



7. Practicing Sub queries (Nested, Correlated).
8. Practicing on Triggers - creation of trigger, Insertion using trigger.
9. Procedures- Creation of Stored Procedures.
10. Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

12. Brief Description of self learning / E-learning component

https://nptel.ac.in/courses/106105175

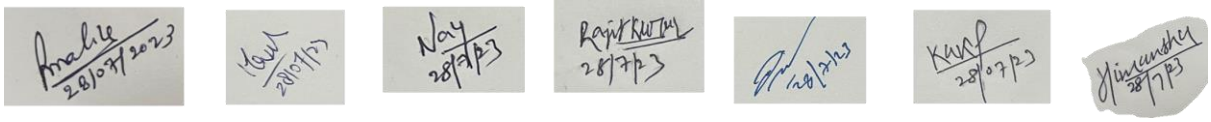
https://nptel.ac.in/courses/106105175

https://nptel.ac.in/courses/106105175

13. Books Recommended

1. Data base Management Systems, Raghurama Krishnan, Johannes Gehrke, TATA McGrawHill 3rd Edition (2014).

2. A. Silberschatz, H.F. Korth and S. Sudharshan, "Database System Concepts" , Fifth Edition, Tata McGraw Hill, New Delhi, 2006.



1. Course Name	Big Data Analytics	**L**	**T**		**P**	
2. Course Code	17200704	**4**	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This course gives an overview of Big Data, i.e., storage, retrieval, and processing of big data. In addition, it also focuses on the "technologies", i.e., the tools/algorithms that are available for storage, processing of Big Data. It also helps a student to perform a variety of "analytics" on different data sets and to arrive at positive conclusions.

8. Course Objectives:

1. Understand the Big Data Platform and its use cases!
2. Provide an overview of Apache Hadoop
3. Provide HDFS Concepts and Interfacing with HDFS
4. Exposure to Data Analytics with R.

9. Course Outcomes (COs):

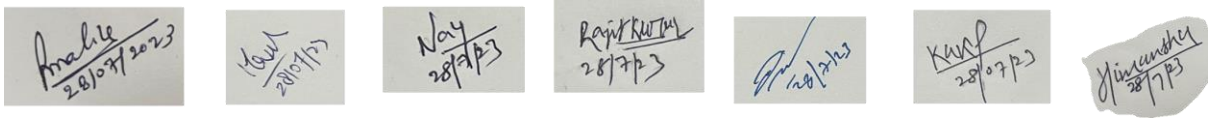
After completing the course, students are expected to be able:

1. Understand Big Data and its analytics in the real world.
2. Analyze the Big Data framework like Hadoop and NOSQL to efficiently store and process Big Data to generate analytics.
3. Design of Algorithms to solve Data Intensive Problems using Map Reduce Paradigm
4. Design and Implementation of Big Data Analytics using pig and spark to solve data intensive problems and to generate analytics.
5. Use the acquired skillsets in developing innovative projects for the benefit of society.

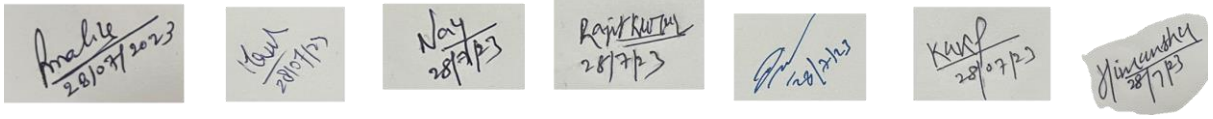
10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Introduction to Big Data**

Distributed File System – Big Data and its importance, Characteristics of Big Data, Limitation of Conventional Data Processing Approaches, Need of big data frameworks, big data analytics, Limitations of Big Data and Challenges, Big data applications.



Unit – 2	Number of lectures = 15	Title of the unit: Big Data in Businesses and Technologies for Handling Big Data
big data in marketing, analytics, retail, hospitality, consumer good, defense etc. Introduction to Hadoop, functioning of Hadoop, Cloud computing (features, advantages, applications) etc.		
Unit – 3	**Number of lectures = 15**	**Title of the unit: Hadoop Ecosystem Components**
Schedulers- Fair and Capacity, Hadoop 2.0 Vs Hadoop 3.0 and its new features. Hadoop Cluster Setup – SSH & Hadoop Configuration –HDFS Administering – Monitoring & Maintenance.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Hadoop MapReduce**
Phases in MapReduce Framework - Anatomy of MapReduce Job run - Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce: Types and Formats, Map Reduce Features. Understanding Basic Map Reduce Program (WordCount program): The Driver Code - The Mapper class - The Reducer class.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/106104189 https://www.youtube.com/watch?v=tcVXPf5TIYo https://www.coursera.org/articles/big-data-analytics		
12. Books Recommended		
1. Boris lublinsky, Kevin t. Smith Alexey Yakubovich, “Professional Hadoop Solutions”. Wiley, 2015. 2. Chris Eaton, Dirk Deroos et al., “Understanding Big Data”, McGraw Hill, 2010. 3. Tom White, “HADOOP”: The definitive Guide”, O Reilly 2012. 4. Srinath Perera, Thilina Gunarathne, "Hadoop MapReduce Cookbook", PACKT publishing, 2013		



14. Name of the Department: Mathematics						
15. Course Name	Big Data Analytics Lab	**L**	**T**		**P**	
16. Course Code	17200705	**0**	0		4	
17. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
18. Pre-requisite (if any)		**19. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()
20. Total Number of Lectures, Tutorials, Practical						
Lectures = 0		**Tutorials =0**	**Practical = 60**			

21. Course Description:

Big data analytics is the use of advanced analytic techniques against very large, diverse data sets that include structured, semi-structured and unstructured data.

22. Course Objectives:

1. To have knowledge on the statistical techniques for Big data Analytics.
2. To acquire understanding in mining data streams.
3. Enable the students to know about clustering techniques.
4. To learn about Hadoop map, Reduce programming.

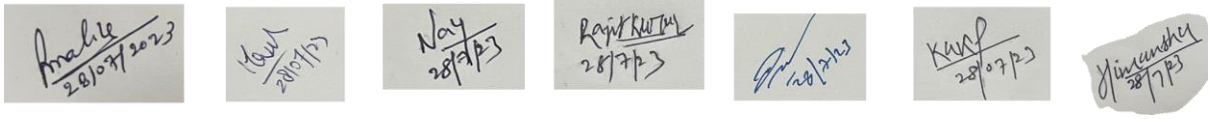
23. Course Outcomes (COs):

Upon completion of the course, the students should be able to:

1. Configure Hadoop and perform File Management Tasks
2. Apply MapReduce programs to real-time issues like word count, weather dataset and sales of a company
3. Critically analyze huge data sets using Hadoop distributed file systems and MapReduce
4. Apply different data processing tools like Pig, Hive and Spark.
5. By learning through experiments innovate, new idea to implement for Research and development.

24. Detailed contents

1. Implement the following using Map- Reduce
a. Matrix multiplication
b. Sorting
c. Indexing
2. Distributed Cache & Map Side Join, reduce side Join Building and Running a Spark Application Word
count in Hadoop and Spark Manipulating RDD
3. Implementation of Matrix algorithms in Spark Sql programming
4. Implementing K-Means Clustering algorithm using Map-Reduce
5. Implementing any one Frequent Item set algorithm using Map-Reduce
6. Create A Data Pipeline Based on Messaging Using PySpark and Hive



7. Develop a MapReduce program to calculate the frequency of a given word in agiven file.
8. Develop a MapReduce program to find the maximum temperature in each year.
9. Install and Run Pig then write Pig Latin scripts to sort, group, join, project, and filter your data.
10. Install and Run Hive then use Hive to create, alter, and drop databases, tables, views, functions, and indexes.
11. Solve some real-life big data problems.
12. Develop a program to calculate the maximum recorded temperature by year for the weather dataset in Pig Latin

25. Brief Description of self-learning / E-learning component

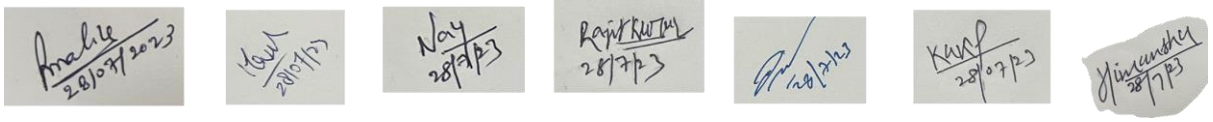
https://nptel.ac.in/courses/106104189

https://www.youtube.com/watch?v=rvJgArru8dI

https://nptel.ac.in/courses/106106142

26. Books Recommended

1. Tom White, Hadoop: The Definitive Guide= Fourth Edition, O'reilly Media, 2015.
2. Glenn J. Myatt, Making Sense of Data, John Wiley & Sons, 2007 Pete Warden, Big Data Glossary, O'Reilly, 2011.
3. Michael Berthold, David J.Hand, Intelligent Data Analysis, Spingers, 2007.
4. Chris Eaton, Dirk DeRoos, Tom Deutsch, George Lapis, Paul Zikopoulos, Uderstanding Big Data Analytics for Enterprise Class Hadoop and Streaming Data, McGrawHill Publishing, 2012.
5. AnandRajaraman and Jeffrey David UIIman, Mining of Massive Datasets Cambridge University Press, 2012.



1. Name of the Department: Mathematics						
2. Course Name	Introduction to Artificial Intelligence	**L**	**T**		**P**	
3. Course Code	17200706	2	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

8. Course Description:

Artificial Intelligence deals with the designing and building of intelligent agents that receive percepts from the environment and take actions that affect that environment. This course is designed to develop the topics of Constraint satisfaction problems, Greedy best first search, Minimax algorithm, Alpha-Beta pruning.

9. Course Objectives:

1. To Solve problems using informed and uninformed search strategies.
2. To Compare various Knowledge Representation Logic using scripts and frames.
3. To Comprehend and analyze the different types of learning.
4. Gain a historical perspective of Artificial intelligence and its foundations.

10. Course Outcomes (COs):

After completing this course students will be able to:

1. Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.
2. Understand and implement search and adversarial algorithms.
3. Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.
4. Explore the fundamental concepts of Artificial intelligence.
5. Use the acquired skillsets in developing innovative data mining projects for the benefit of society.

11. Unit wise detailed content

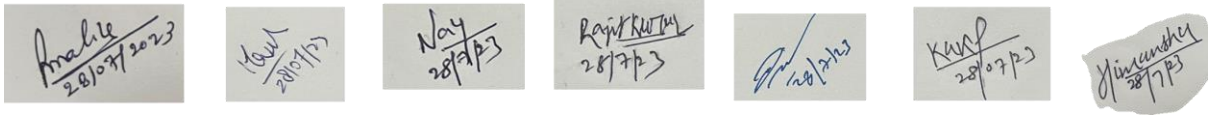
Unit-1	**Number of lectures = 8**	**Title of the unit: Introduction**

Introduction to AI, The AI problems, AI technique, philosophy and development of Artificial intelligence, Production system characteristics, Searching, Uniformed search strategies – Breadth first search, depth first search.

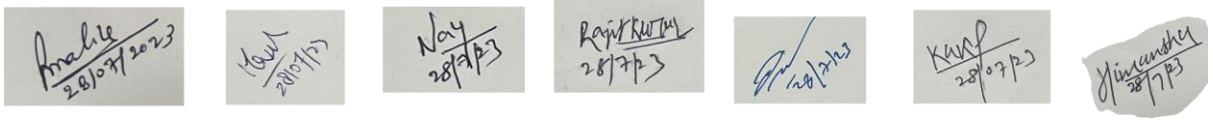
Unit – 2	**Number of lectures = 8**	**Title of the unit:** Local search algorithms

Generate and Test, Hill climbing, simulated annealing search, Constraint satisfaction problems, Greedy best first search, A* search, AO* search, Uninformed and informed search techniques.

Unit – 3	**Number of lectures = 8**	**Title of the unit: Knowledge representation and reasoning**



Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules-based deduction systems. Reasoning under uncertainty, review of probability.		
Unit – 4	**Number of lectures = 6**	**Title of the unit: Communicating, perceiving, and acting**
Communication as Action- Syntactic Analysis- Augmented Grammars- Semantic Interpretation- Ambiguity.		
12. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/106102220 https://nptel.ac.in/courses/112103280 https://nptel.ac.in/courses/106105078		
13. Books Recommended		
1. W. Patterson, 'Introduction to Artificial Intelligence and Expert Systems', Prentice Hall of India, 2003. 2. N. J. Nilsson, Artificial Intelligence A New Synthesis, Morgan Kauffman Pub 3. J Giarratano and G Riley, Expert Systems - Principles and Programming, Thomson Publishing House. 4. Deepak Khemani, Artificial Intelligence, Tata Mc Graw Hill Education, 2017. 5. N. P. Padhy: Artificial Intelligence and Intelligent Systems, Oxford Higher Education, Oxford University Press, 2005.		



1. Name of the Department: Mathematics						
2. Course Name	Fuzzy Logic & Fuzzy Decision Making	**L**	**T**		**P**	
3. Course Code	17200707	2	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. re-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

8. Course Description:

This course provides the fundamentals of classical set theory and fuzzy set theory. The decomposition theorems of fuzzy sets and the extension principle will be introduced, as well as the use of nonlinear integrals as aggregation tools to deal with fuzzy data. As an indispensable tool in fuzzy decision making, ranking and ordering fuzzy quantities will be discussed.

9. Course Objectives:

To provide an understanding of the basic mathematical elements of the theory of fuzzy sets.

To provide an emphasis on the differences and similarities between fuzzy sets and classical sets theories.

To cover fuzzy logic inference with emphasis on their use in the design of intelligent or humanistic systems.

To provide a brief introduction to fuzzy arithmetic concepts.

10. Course Outcomes (COs):

Objectives: Upon successful completion of this course, students should be able to:

1. Understand basic knowledge of fuzzy sets and fuzzy logic,
2. Apply fuzzy inferences,
3. Apply fuzzy information in decision-making,
4. Appreciate the theory of possibility on the basis of evidence**.**
5. Demonstrate fuzzy set theory models and their daily life application.

11. Unit wise detailed content

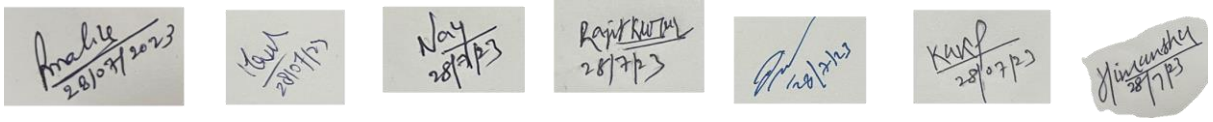
Unit-1	**Number of lectures = 8**	**Title of the unit: Introduction to Fuzzy Sets**

Basic definitions, α -level sets, comparison with classical (crisp) sets, Types of fuzzy sets, extension principle, Fuzzy complement, t-norms, t-co-norms, a combination of operations, aggregation operations.

Unit – 2	**Number of lectures = 8**	**Title of the unit: Crisp versus fuzzy relation**

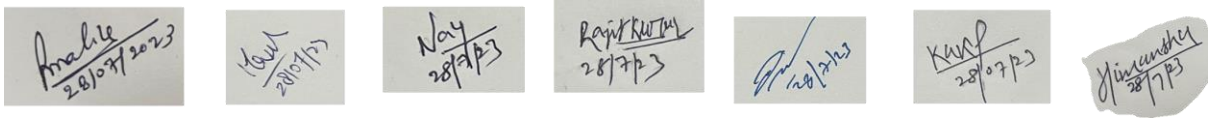
Crisp versus fuzzy relation, projections and cylindrical extensions, binary fuzzy relations, binary relations on a single set, fuzzy equivalence relations, and fuzzy compatibility.

Unit – 3	**Number of lectures = 8**	**Title of the unit: Fuzzy Logic**



An overview of classical logic, multi valued logic, fuzzy propositions, fuzzy quantifiers, and linguistic hedges, Inference from conditional fuzzy propositions, Inference from conditional and qualified propositions.		
Unit – 4	**Number of lectures = 6**	**Fuzzy Linear Programming**
Multicriteria decision making, fuzzy ranking method, fuzzy linear programming. Methods of de-fuzzyfication.		
12. Brief Description of self-learning / E-learning component		
https://www.youtube.com/watch?v=IZWTduVCrf8 https://www.youtube.com/watch?v=w-hHe5g41u4 https://nptel.ac.in/courses/108104157		
13. Books Recommended		

1. George J. Klir and Bo Yuan, Fuzzy Sets and Fuzzy Logic: Theory and Applications, Prentice Hall of India, New Delhi.
2. H.J. Zimmermann, Fuzzy Set Theory & its Applications, Allied Publishers Ltd. New Delhi.
3. Timothy J. Ross, Fuzzy Logic with Engineering Applications, McGraw Hills inc. New Delhi



1. Name of the Department: Mathematics						
2. Course Name	Introduction of Stochastic Process	**L**	**T**		**P**	
3. Course Code	17200708	**2**	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 30	**Tutorials = 0**	**Practical = 0**

8. Course Description:

This course prepares students to a rigorous study of Stochastic Processes. Towards this goal, we cover elements of stochastic process and emphasize the applications to stochastic processes. Main topics are introduction to stochastic processes, examples such as Markov chains, Poison Process, birth and death process and Applications of stochastic processes in queuing & reliability.

9. Course Objectives:

1. Learn stochastic process and its large variety from introduction to an intermediate level of application knowledge.
2. Learn the stochastic processes in queues and understand different queue models.
3. Understand stochastic processes on depth and find avenues for further research.

10. Course Outcomes (COs):

The students will be able to:

1. Apply Markovian model stochastic processes and obtain solutions especially in the field of engineering
2. Derive new queue models to provide better solutions.
3. Find solutions for the untoward happening using the knowledge on reliability theory.
4. Indulge in strong research to get solutions in all walks of life since everything is probabilistic.
5. Demonstrate Stochastic models and their daily life application.

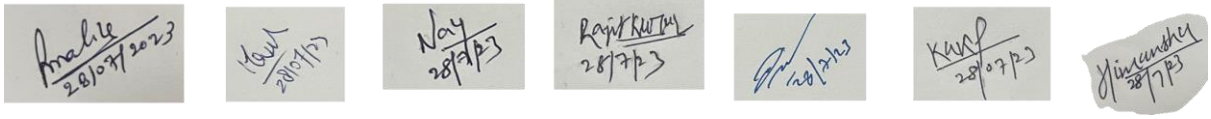
11. Unit wise detailed content

Unit-1	**Number of lectures = 08**	**Title of the unit: Stochastic Process**

Introduction of Stochastic Processes: definition, classification and examples, classification of random processes, according to state space and parameter space, types of SPs, elementary problems.

Unit – 2	**Number of lectures = 06**	**Title of the unit: Markov Chains**

Markov Chains: definition and examples, Transition probability matrix (P), order of a Markov chain, classification of states, Stationary distribution



Unit – 3	Number of lectures = 08	Title of the unit: Markov Process
Poisson Process: Introduction, postulates, properties and related distributions, Simple birth-process, Simple death-process, Simple birth-death process.		
Unit – 4	**Number of lectures = 08**	**Title of the unit: Stochastic Processes in Queuing**
Queuing models, birth and death processes in queuing theory, Markovian queuing models, non-Markovian queuing models.		

12. Brief Description of self-learning / E-learning component

https://www.youtube.com/watch?v=KUDhXlnr-gU

https://nptel.ac.in/courses/111102014

https://nptel.ac.in/courses/111103022

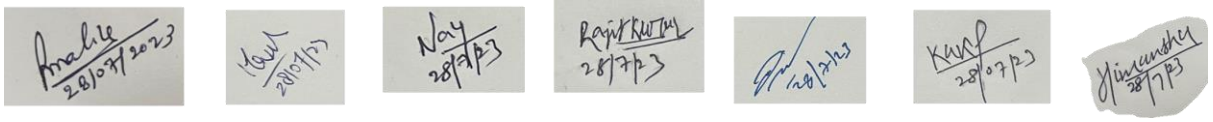
13. Books Recommended

1. J. Medhi, Stochastic Processes, New Age International Publishers, 2009

2. Bailey, Norman T. (1965): The Elements of Stochastic Processes, John Wiley & Sons, Inc., New York.

3. E. Balagurusami, Reliability Engineering, Tata McGraw Hill, New Delhi, 1984.

4. L. S. Srinath, Reliability Engineering, Affiliated East West Press, New Delhi, 1991.



1. Name of the Department: Mathematics						
2. Course Name	Numerical Method	**L**	**T**		**P**	
3. Course Code	17200710	2	0		0	
4. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even ()	Odd (✓)	Either Sem ()	Every Sem ()

7. Total Number of Lectures, Tutorials, Practical		
Lectures = 30	**Tutorials = 0**	**Practical = 0**

8. Course Description:

This course analyzed the basic techniques (direct and iterative methods) for the efficient numerical solution of problems in science and engineering. Topics s covered are: Number representation and errors, Polynomials, Locating roots of equations, Solution of nonlinear equations, Interpolation and approximation, Numerical differentiation, Numerical integration, Systems of linear equations, Solution of differential equations.

9. Course Objectives:

Many applications in engineering, physics, geology and other specifications containing complicated problems that will require one of the numerical methods to be solved. In this course students will learn the classification of many complicated problems and the suitable numerical methods for obtaining an approximated solution to these problems with desired accuracy.

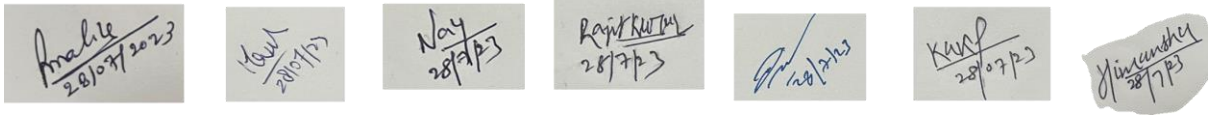
10. Course Outcomes (COs):

On completion of this course, the students will be able:

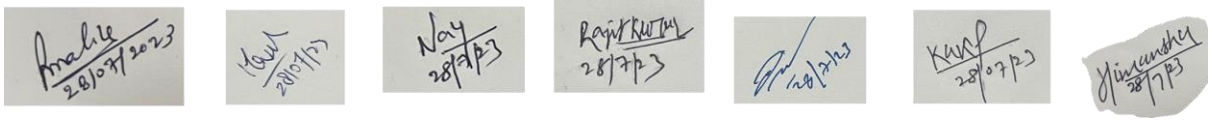
1. To demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions to otherwise intractable mathematical problems.
2. To derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.
3. To analyse and evaluate the accuracy of differentiation and integrations using numerical methods.
4. To evaluate the system of linear equations and ordinary differential equations using numerical methods.
5. To use the acquired skillsets in developing innovative projects for the benefit of society.

11. Unit wise detailed content

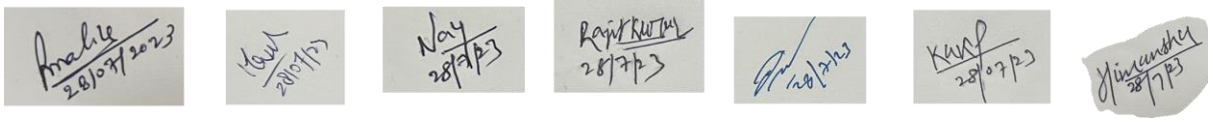
Unit-1	**Number of lectures = 8**	**Title of the unit: Errors & Solution of transcendental and algebraic equations**



Representations of numbers: Round off error, truncation error, significant error, error in numerical computations. Bisection, secant, Regula Falsi, fixed-point, Newton-Raphson, Graffe's methods.		
Unit-2	**Number of lectures = 8**	**Title of the unit: Interpolation & Approximation**
Difference schemes, and interpolation formulas using differences. Lagrange and Newton interpolation. Hermite interpolation. Divided differences.		
Unit-3	**Number of lectures = 8**	**Title of the unit: Numerical Differentiation & Numerical Integration**
Numerical differentiation, Methods based on interpolations, Methods based on finite differences, Numerical integration: Trapezoidal, and Simpson's.		
Unit- 4	**Number of lectures = 6**	**Title of the unit: Solution of linear equations and Numerical Solution of IVP(ODE)**
Direct methods - Gauss elimination, Gauss-Jordan elimination, LU decomposition. Iterative methods - Jacobi, Gauss-Siedel; The algebraic eigenvalue problem: Jacobi's method, Power method.		
Brief Description of self-learning / E-learning component:		
https://archive.nptel.ac.in/content/storage2/courses/122104019/numerical-analysis/kadalbajoo/lec1/fnode6.html http://www.digimat.in/nptel/courses/video/111107105/L26.html http://nitttrc.edu.in/nptel/courses/video/122102009/L35.html		
12. Books Recommended		
1. Richard L. Burden and J. Douglas Faires, Numerical Analysis, Brookes Cole 2004. 2. M.K. Jain, S.R.K. Iyengar and R.K.Jain, Numerical Methods for Scientific and Engineering Computation, New Age international Publishers, New Delhi, India, 2003. 3. Chapra, S. and R. Canale, Numerical Methods for Engineers. New York: McGraw Hill 1998. 4. R. K. Gupta. Numerical Methods: Fundamentals and Applications, Cambridge university press; first edition, 2019.		



Semester-VIII



1. Name of the Department: Mathematics						
2. Course Name	Deep Learning Techniques	**L**	**T**		**P**	
3. Course Code	17200801	4	0		0	
4. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
5. Pre-requisite (if any)		**6. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()
7. Total Number of Lectures, Tutorials, Practical						
Lectures = 60		**Tutorials = 0**	**Practical = 0**			

8. Course Description:

This course introduces fundamentals of feedforward networks, Optimization for Training Deep Models and Convolutional Networks, Recurrent Neural Networks (RNN), Some Popular Deep networks and Applications, Applications of deep learning.

9. Course Objectives:

1. To understand the theoretical foundations, algorithms and methodologies of feedforward networks.
2. To design and develop an application using specific deep learning models
3. To provide the practical knowledge in handling and analyzing real world applications.

10. Course Outcomes (COs):

At the end of the course students should be able:

1. To obtain a concept of deep learning and its advantages.
2. To implement different deep learning algorithms
3. To design the test procedures to assess the efficacy of the developed model.
4. To explore the research domain of deep learning.
5. To demonstrate deep learning models and their daily life application.

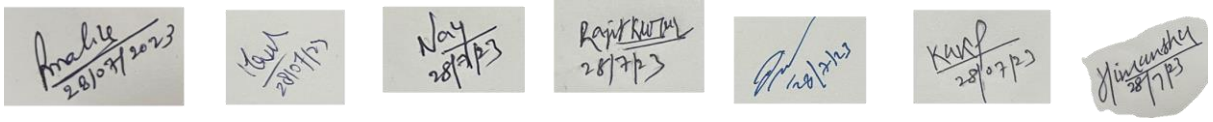
11. Unit wise detailed content

Unit-1	**Number of lectures = 15**	**Title of the unit:** Introduction:

Extracting meaning from data, expert system, learning algorithms, overfitting and underfitting, regularization, hyperparameters and validation sets, estimator, bias and variance, Bayesian statistics, supervised learning, unsupervised learning, Stochastic Gradient Descent, building a machine learning algorithm, challenges motivating Deep Learning.

Unit – 2	**Number of lectures = 15**	Title of the unit: Fundamentals of feedforward networks

Single-layer and multilayer feedforward networks, Neural Network Graphs, activation functions, deep feedforward networks, gradient-based learning, Back-propagation algorithm and other differentiation algorithms, Regularization for deep learning, Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Early Stopping, Sparse.



Unit – 3	Number of lectures = 15	Title of the unit: Optimization for Training Deep Models and Convolutional Networks
How Learning Differs from Pure Optimization, Challenges in Neural Network Optimization, Basic Algorithms, Parameter Initialization Strategies, Algorithms with Adaptive Learning, Rates, Approximate Second-Order Methods, Batch Normalization, The Convolution Operation, Pooling, Variants of the Basic Convolution Function, Structured Outputs.		
Unit – 4	**Number of lectures = 15**	**Title of the unit:** Sequence Modelling, Recurrent Neural Networks (RNN)
Unfolding Computational Graphs, RNNs, Bidirectional RNNs, LSTM, Stochastic Encoders and Decoders, Some Popular Deep networks and Applications: Generative Adversarial Networks, VGG, Applications of deep learning.		

12. Brief Description of self-learning / E-learning component

https://nptel.ac.in/courses/106106184

https://nptel.ac.in/courses/106106184

https://nptel.ac.in/courses/106106224

13. Books Recommended

1. Duda, R.O., Hart, P.E., and Stork, D.G. Pattern Classification. Wiley-Interscience. 2nd Edition. 2001.

2. Russell, S. and Norvig, N. Articial Intelligence: A Modern Approach. Prentice Hall Series in Articial Intelligence. 2003.

3. Nikhil Buduma, Nicholas Locascio, “Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms”, O'Reilly Media, 2017.

4. I. Goodfellow, Y. Bengio and A. Courville, Deep Learning, The MIT Press, 2017.

5. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer, 2018

1. Course Name	Introduction to Decision modelling	**L**	**T**		**P**	
2. Course Code	17200802	**4**	0		0	
3. Type of Course (use tick mark)		**Core (✓)**	**DSE ()**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 60	**Tutorials = 0**	**Practical = 0**

7. Course Description:

This course will show you how to analyze a wide array of business planning and decision problems involving data with the help of spreadsheet models. We provide a framework for quantitative decision-making, non-cooperative game theory, Monte-Carlo Simulation, effective resource allocation and economic efficiency.

8. Course Objectives:

From this course, students should:

1. become aware of the scope of management problems that can be addressed with models.
2. be able to identify the essential conceptual structure of a decision/planning problems.
3. recognize the types of modeling tools most adapted to a given situation;

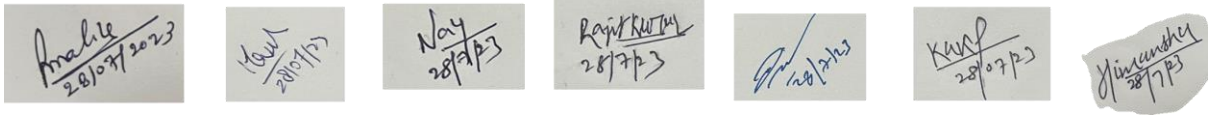
9. Course Outcomes (COs):

After completing this course, students will have learnt:

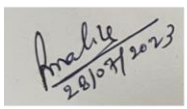
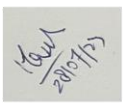
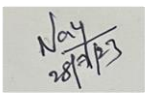
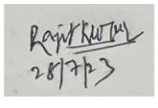
1. How to build models by payoff matrix.
2. New techniques to solve real-life complex problems.
3. To solve real life problems through game theory.
4. To measure the effect of change and discover techniques to improve your decision-making process using Monte-Carlo Simulation.
5. Demonstrate decision-making models and their daily life application.

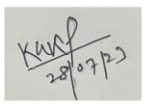
10. Unit wise detailed content

Unit – 1	**Number of lectures = 15**	**Title of the unit: Introduction of Decision Analysis:**
Introduction of payoff Matrix, Decision Making Under Risk, Value of Information, Probability Concepts.		
Unit – 2	**Number of lectures = 15**	**Title of the unit:** Decision Problem with Experimentation



Bayes Theorem, Decision Tree, Decision Problem with Experimentation, Decision Problem with Experimentation (Contd.), Decision Problem Example.		
Unit – 3	**Number of lectures = 15**	**Title of the unit:** Non-Cooperative Game Theory:
Introduction to Game Theory, Extensive Form Games, Strategic Form Games, Dominant Strategy Equilibrium, Pure Strategy Nash Equilibrium, Mixed Strategy Nash Equilibrium, Sperner's Lemma, Fixed Point Theorem and Existence of Nash Equilibrium, Matrix Games (Two Players Zero Sum Games), Bayesian Games, Subgame Perfect Equilibrium.		
Unit – 4	**Number of lectures = 15**	**Title of the unit: Simulation models**
Introduction to Simulation: Discrete-Event and Monte-Carlo Simulation, Pseudo random Numbers, Simulation Examples, Generation of Random Variates.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/110105082 https://www.youtube.com/watch?v=6OeaM3MZS8E https://www.youtube.com/watch?v=6OeaM3MZS8E https://in.coursera.org/learn/simulation-models-for-decision-making		
12. Books Recommended		
1. Y. Narahari. Game Theory and Mechanism Design. World Scientific Publishing Company Pte. Limited, 2014, ISSN: 978-9814525046. 4. T. 2. Roughgarden, Twenty Lectures on Algorithmic Game Theory, Cambridge University Press, 2016, ISSN: 978-1316624791. 2. Hillier F.S. and Lieberman G.J., Introduction to Operation Research, McGraw Hill. 3. Banks, J.S., Carson, Nelson B.L. and Nicol D.M., Discrete Event System Simulation, Pearson Education International Series. 4. Law A.M. and Kelton W.D., Simulation Modeling and Analysis, McGraw Hill International - Industrial Engg. Series.		




1. Course Name	Introduction to Financial Mathematics	**L**	**T**		**P**	
2. Course Code	17200803	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

Investment mathematics emphasizing the time value of money, rates of return for investment cash-flow sequences, utility functions, stochastic processes, mean-variance analysis, portfolio selection, hedging strategies, the capital assets pricing model, and the Black-Scholes theory of options. This course will also cover some of the topics covered on the Society of Actuaries' actuarial exams SOA/FM and SOA/MFE.

8. Course Objectives:

This course's objectives include teaching students about the mathematical treatment of risk-neutral valuation, arbitrage, options, stochastic modelling, and derivatives.

9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1.To make students aware of the economic aspects of consumer choice employing optimization
2. To make students understand how to use optimization concept in case of economic operations based on production and cost concept.
3. To make students conversant about various structure and output decision with the help of principle of optimization.
4. To use the acquired skillsets in developing innovative projects for the benefit of society.
5. To demonstrate econometric models and their daily life application.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 8**	**Title of the unit: Introduction of Financial Mathematics**

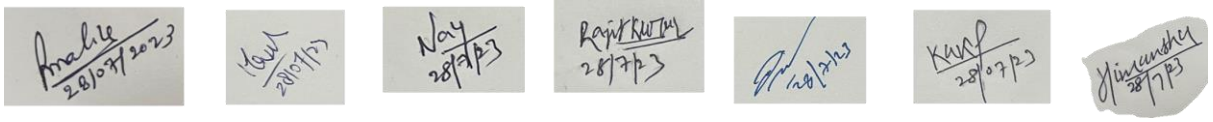
Introduction to financial mathematics, Progressions and Series, Growth and Decay Curves, Statistical Measures, Problem-Solving on Mathematical Functions and Statistical Measures, interest and Interest Rate, Time Value of Money, and Simple Discount.

Unit – 2	**Number of lectures = 08**	**Title of the unit:** Discrete Compounding

Introduction to Discrete Compounding and Discrete Payments, Equal Payment Series and Gradient Series Factors, Geometric Gradient Series Factors.

Unit – 3	**Number of lectures = 6**	**Title of the unit: Production and Cost Concept:**

Concept of Production function, Production with one and two variable inputs. Cobb Douglas Form. Law of Diminishing Marginal Return and Marginal Rate of Technical Substitution. Concept of Return to Scale.		
Unit – 4	**Number of lectures = 8**	**Title of the unit:** Theory of Market:
Marginal Cost, Marginal revenue and Profit Maximization by a competitive Firm, Short run and long run output decision, Monopoly market features and monopolist's output decision.		
11. Brief Description of self-learning / E-learning component		
https://nptel.ac.in/courses/112107260 https://www.youtube.com/watch?v=Y_0PeI6DEVI https://archive.nptel.ac.in/courses/111/103/111103126/		
12. Books Recommended		
1. R S Pindyck, D L Rubinfeld and PL Mehta, Microeconomics, (Pearson Education) 2. A Koutsoyanis, Modern Microeconomics, (Macmilan Education) 3. P Tandon, A Textbook of Microeconomic Theory, (Sage Publication). 4. S A Greenlaw and D Shapiro, Principle of Microeconomics, Open Stax (Rice University). 5. Marek Capinski and Tomasz Zastawniak, "Mathematics for Finance", Springer. 6. Ambad Nazri Wahidudin, "Financial Mathematics and its Applications", Ventus Publishing ApS		



1. Course Name	Data Mining and Data Warehousing	**L**	**T**		**P**	
2. Course Code	17200804	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

Data mining is a class of analytical techniques that examine a large amount of data to discover new and valuable information. This course is designed to introduce the core concepts of data mining, its techniques, implementation, benefits, and outcome expecta-tions from this new technology. This course introduces data warehousing and data mining techniques and their software tools. Topics include data warehousing, association analysis, classification, cluster-ing, numeric prediction, and selected advanced data mining topics.

8. Course Objectives:

The objective of this course is to:

1. Introduce data mining principles and techniques.
2. Introduce data mining as a cutting-edge business intelligence tool.
3. Develop and apply critical thinking, problem solving and decision-making skills.
4. Introduce the concepts of Data Warehousing, difference between database and data warehousing.
5. Describe and demonstrate basic data mining algorithms, methods, and tools.

9. Course Outcomes (COs):

After completing the course, students are expected to be able to:

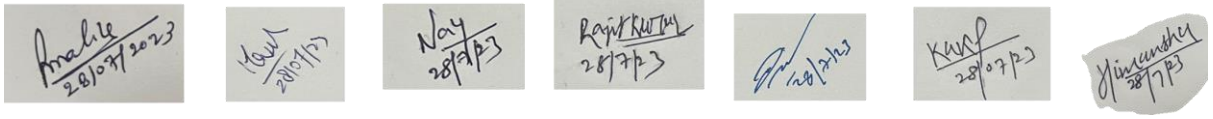
1. Design a data warehouse or data mart to present information needed by the and can be utilized for managing clients.
2. Design and implement a quality data warehouse or data mart effectively and administer the data resources in such a way that it will truly meet management's requirements.
3. Evaluate standards and new technologies to determine their potential impact on your information resource for a large complex data warehouse/data mart.
4. Use data mining tools for projects and to build reliable products as per demand.

10. Unit-wise detailed content

Unit – 1	**Number of lectures = 8**	**Title of the unit: Introduction**

Motivation for Data Mining, Data Mining-Definition & Functionalities, Data Processing, Form of Data Preprocessing, Data Cleaning: Missing Values, Noisy Data (Binning, Clustering, Regression, Computer and Human inspection), Inconsistent Data, Data Integration and Transformation.

Unit – 2	**Number of lectures = 8**	**Title of the unit: Concept Description and Classification**
Data Generalization, Analytical Characterization, Analysis of attribute relevance, Mining Class comparisons, and Statistical measures in Large Databases. Graph Displays of Basic Statistical class Description.		
Unit – 3	**Number of lectures = 6**	**Title of the unit: Clustering and Applications**
Cluster analysis, Types of Data in Cluster Analysis, Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical Methods, and Density–Based Methods.		
Unit – 4	**Number of lectures = 8**	**Title of the unit: Data Warehousing:**
Overview, Definition, Delivery Process, Difference between Database System and Data Warehouse, Multi-Dimensional Data Model, Data Cubes, Stars, Snowflakes, Fact Constellations, Concept hierarchy, Process Architecture, 3 Tier Architecture.		
11. Brief Description of self-learning / E-learning component		
https://onlinecourses.nptel.ac.in/noc21_cs06/preview https://archive.nptel.ac.in/courses/106/105/106105174/ https://onlinecourses.swayam2.ac.in/cec20_cs12/preview		
12. Books Recommended		
1. Ian H. Witten and Eibe Frank, Data Mining: Practical Machine Learning Tools and Techniques (Second Edition), Morgan Kaufmann, 2005. 2. H.Dunham, "Data Mining: Introductory and Advanced Topics" Pearson Education. 3. Sam Anahory, Dennis Murray, "Data Warehousing in the Real World: A Practical Guide for Building Decision Support Systems, Pearson Education. 4. Jiawei Han, Micheline Kamber," Data Mining Concepts & Techniques" Elsevier. 5. Mallach," Data Warehousing System", McGraw –Hill.		



1. Course Name	Mathematical Economics	**L**	**T**		**P**	
2. Course Code	17200805	2	0		0	
3. Type of Course (use tick mark)		**Core ()**	**DSE (✓)**	**AEC ()**	**SEC ()**	**OE ()**
4. Pre-requisite (if any)		**5. Frequency (use tick marks)**	Even (✓)	Odd ()	Either Sem ()	Every Sem ()

6. Total Number of Lectures, Tutorials, Practical

Lectures = 30	**Tutorials = 0**	**Practical = 0**

7. Course Description:

The principal aim of this course is to extend your facility with those methods of mathematics needed to pursue economic analysis at a more advanced level. By the end of the course, you should have extended your knowledge to include the technique of optimization under inequality constraints, game theory economic models, and differential and difference equations.

8. Course Objectives:

1. To familiarize the basic concepts and theories of economics, elementary statistics and mathematics
2. To learn the basics of Optimization methods and economic analysis.
3. To understand the concepts such as Input-output analysis, structure of an economy etc.
4. To learn applications of Game theory in economics.

9. Course Outcomes (COs):

After completing the course, students are expected to be able:

1. How to build models by expressing words in symbols, numbers and equations
2. New techniques to solve complex problems.
3. To measure the effect of change and discover techniques to improve your decision-making process.
4. Economic dynamics and solve problems through adjustments with time.
5. A new dimension of scientific, logical and critical thinking that will assist your mind to solve personal, professional and social problems and guide you to take wise decisions.

10. Unit wise detailed content

Unit – 1	**Number of lectures = 8**	**Title of the unit: Introduction to Optimization**

Optimization methods and economic analysis, application of constrained and unconstrained optimization methods to consumer and producer behaviour production function.

Unit – 2	**Number of lectures = 8**	**Title of the unit: Linear Programming**

Linear programming, primal, dual, graphic method, simplex method, application to production and diet problems, Non-linear programming, Hawkin-Simon conditions, method and applications.

Unit – 3	**Number of lectures = 6**	**Title of the unit: Input–output analysis**

Input–output analysis, structure of an economy, assumptions, technical coefficient, outputs and price determination, static and dynamic input-output analysis.

Unit – 4	Number of lectures = 8	Title of the unit: Game Theory and difference equation
Game theory, basic concepts, two-person zero-sum game, saddle point, examples of co-operative and non-co-operative games, Difference and differential equations, first and second-order linear differential.		
11. Brief Description of self-learning / E-learning component		
https://onlinecourses.swayam2.ac.in/ugc19_hs41/preview https://www.digimat.in/nptel/courses/video/109103188/L01.html https://www.youtube.com/watch?v=V3mqM3n9B_4&feature=emb_imp_woyt https://nptel.ac.in/courses/109103188		
12. Books Recommended		
1. A.C. Chiang: Fundamental Methods of Mathematical Economics, McGraw Hill. 2. J. Henderson and R.E. Quandt: Micro Economic Theory, Tata McGraw Hill. 3. M.D.Intrilligator: Mathematical Optimization and Economic Theory, Prentice Hall.		

